

title: "MasterControl -Lead Scoring & Conversion Optimization" subtitle: "Group Modeling Notebook: Feature Engineering, Model Selection & Business Impact" author: "Thomas Beck · Max Ridgeway · Astha KC" date: "Spring 2026" format: html: embed-resources: true theme: flatly highlight-style: tango toc: true toc-depth: 3 toc-location: left number-sections: true df-print: paged code-fold: true code-tools: true pdf: documentclass: article geometry: - top=0.6in - bottom=0.6in - left=0.6in - right=0.6in fontsize: 10pt toc: true toc-depth: 2 number-sections: true colorlinks: true code-fold: true fig-pos: "H" keep-tex: false execute: echo: true warning: false message: false fig-width: 7.5 fig-height: 4.5 fig-dpi: 200 out-width: "100%" editor_options: chunk_output_type: inline

```
In [1]: #| label: setup
#| include: false

import subprocess, sys, warnings, time, re, multiprocessing
import numpy as np
import pandas as pd

warnings.filterwarnings('ignore')
np.random.seed(42)
START_TIME = time.time()

# Reserve 2 cores for other tasks
N_JOBS = max(1, multiprocessing.cpu_count() - 2)
```

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In [2]: #| label: dependencies
#| include: false

def install_if_missing(package_name, import_name=None, pip_name=None):
    import_name = import_name or package_name.lower()
    pip_name = pip_name or import_name
    try:
        __import__(import_name)
    except ImportError:
        subprocess.check_call(
            [sys.executable, "-m", "pip", "install", pip_name, "-q"],
            stdout=subprocess.DEVNULL, stderr=subprocess.DEVNULL)
```

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for pkg in [
    ("scikit-learn", "sklearn", "scikit-learn"),
    ("pyprojroot", None, None),
    ("CatBoost", "catboost", None),
    ("XGBoost", "xgboost", None),
    ("LightGBM", "lightgbm", None),
    ("SHAP", "shap", None),
    ("matplotlib", None, None),
    ("seaborn", None, None),
    ("tabulate", None, None),
    ("imbalanced-learn", "imblearn", "imbalanced-learn"),
]:
    install_if_missing(*pkg)

```

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In [3]: #| label: imports
        #| include: false

import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
import matplotlib.ticker as mticker
import seaborn as sns
from pathlib import Path
from pyprojroot import here
from types import SimpleNamespace
from tabulate import tabulate

from sklearn.model_selection import (
    train_test_split, StratifiedKFold,
    RandomizedSearchCV, cross_val_predict,
)
from sklearn.preprocessing import (
    StandardScaler, LabelEncoder,
    FunctionTransformer, OrdinalEncoder,
)
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.decomposition import TruncatedSVD
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.base import (
    BaseEstimator, TransformerMixin,
    ClassifierMixin, clone,
)
from sklearn.metrics import (
    roc_auc_score, roc_curve,
    precision_recall_curve, average_precision_score,
    classification_report, confusion_matrix,
    brier_score_loss, log_loss,
    f1_score, precision_score, recall_score,
)
from sklearn.calibration import (
    CalibratedClassifierCV, calibration_curve,
)
from sklearn.ensemble import (
    RandomForestClassifier, GradientBoostingClassifier,
    StackingClassifier, VotingClassifier,

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)
from sklearn.linear_model import LogisticRegression

SMOTE_AVAILABLE = False
try:
    from imblearn.over_sampling import SMOTE
    SMOTE_AVAILABLE = True
except ImportError:
    pass

```

```

In [4]: #| label: configuration
        #| include: false

RANDOM_STATE = 42
CV_FOLDS = 5
N_ITER_SEARCH = 150
TEST_SIZE = 0.10
VAL_SIZE = 0.10

LSA_COMPONENTS = 20
TFIDF_MAX_FEATURES = 500

COST_PER_CALL = 50
VALUE_PER_SQL = 6000

SHAP_BACKGROUND_SAMPLES = 100
SHAP_TEST_SAMPLES = 200

COL_SUCCESS = '#00534B'
COL_RISK = '#F05627'
COL_NEUTRAL = '#95a5a6'
COL_ACCENT = '#2980b9'
COL_GOLD = '#f39c12'
COL_PROFIT = '#27ae60'
COL_LOWVAL = '#e74c3c'
COL_PREMIUM = '#2ecc71'

MODEL_COLORS = {
    'CatBoost': '#1b9e77',
    'XGBoost': '#d95f02',
    'LightGBM': '#7570b3',
    'GradientBoosting': '#e7298a',
    'RandomForest': '#66a61e',
    'StackingEnsemble': '#00534B',
    'VotingEnsemble': '#e6ab02',
}

# PDF-friendly defaults
sns.set_theme(style="whitegrid", context="paper", font_scale=1.1)
plt.rcParams.update({
    'figure.figsize': (7.5, 4.5),
    'figure.dpi': 200,
    'axes.titleweight': 'bold',
    'font.family': 'sans-serif',
    'figure.autolayout': True,
    'savefig.bbox': 'tight',

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    'savefig.pad_inches': 0.1,
})

def print_table(data, caption="", fmt="pipe"):
    if isinstance(data, pd.DataFrame):
        print(f"\n**{caption}**\n" if caption else "")
        print(data.to_markdown(index=False))
    else:
        print(data)

DATA_DIR = here("data")
OUTPUT_DIR = here("output")
CLEANED_DATA_PATH = here(
    "output/Cleaned_QAL_Performance_for_MSBA.csv"
)
RAW_DATA_PATH = here(
    "data/QAL Performance for MSBA.csv"
)
DATA_PATH = (
    CLEANED_DATA_PATH
    if CLEANED_DATA_PATH.exists()
    else RAW_DATA_PATH
)

```

```

In [5]: #| label: catboost-wrapper
#| include: false
#| echo: false

CATBOOST_AVAILABLE = False
try:
    from catboost import CatBoostClassifier as CatBoostRaw

    class SklearnCatBoost(BaseEstimator, ClassifierMixin):
        _estimator_type = "classifier"

        def __init__(
            self, iterations=500, depth=6,
            learning_rate=0.1, l2_leaf_reg=3,
            border_count=64, random_state=42,
            verbose=0, thread_count=1,
        ):
            self.iterations = iterations
            self.depth = depth
            self.learning_rate = learning_rate
            self.l2_leaf_reg = l2_leaf_reg
            self.border_count = border_count
            self.random_state = random_state
            self.verbose = verbose
            self.thread_count = thread_count
            self._model = None

        def __sklearn_tags__(self):
            tags = SimpleNamespace()
            tags.estimator_type = "classifier"
            tags.classifier_tags = SimpleNamespace()
            tags.regressor_tags = None

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tags.transformer_tags = None
tags.input_tags = SimpleNamespace(
    allow_nan=True, pairwise=False,
    one_d_labels=True, two_d_labels=False,
)
tags.target_tags = SimpleNamespace(
    required_y=True, one_d_labels=True,
    two_d_labels=False,
)
return tags

def fit(self, X, y, **kw):
    self._model = CatBoostRaw(
        iterations=self.iterations,
        depth=self.depth,
        learning_rate=self.learning_rate,
        l2_leaf_reg=self.l2_leaf_reg,
        border_count=self.border_count,
        random_state=self.random_state,
        verbose=self.verbose,
        thread_count=self.thread_count,
        allow_writing_files=False,
    )
    self._model.fit(X, y, **kw)
    self.classes_ = np.unique(y)
    return self

def predict(self, X):
    return (
        self._model.predict(X)
        .flatten().astype(int)
    )

def predict_proba(self, X):
    return self._model.predict_proba(X)

@property
def feature_importances_(self):
    return self._model.get_feature_importance()

CATBOOST_AVAILABLE = True
except ImportError:
    pass

XGB_AVAILABLE = False
try:
    from xgboost import XGBClassifier
    XGB_AVAILABLE = True
except ImportError:
    pass

LIGHTGBM_AVAILABLE = False
try:
    from lightgbm import LGBMClassifier
    LIGHTGBM_AVAILABLE = True
except ImportError:

```

```

pass

TARGET_ENCODER_AVAILABLE = False
try:
    from sklearn.preprocessing import TargetEncoder
    TARGET_ENCODER_AVAILABLE = True
except ImportError:
    pass

SHAP_AVAILABLE = False
try:
    import shap
    SHAP_AVAILABLE = True
except ImportError:
    pass

```

Introduction

Business Problem

MasterControl sells quality management software to regulated industries through two product lines: **Mx** (manufacturing execution) and **Qx** (quality management). Leads enter the pipeline as Qualified Activity Leads (QALs) and progress through stages - Disqualified, Recycled, SQL, SQO, or Won.

The core challenge is **conversion efficiency**. Mx converts at roughly 12.7% versus 19.7% for Qx - a 7 percentage-point gap. With a *50costpersalescall* and 6,000 value per SQL conversion, the ability to identify which leads are most likely to convert has direct revenue impact.

Analytic Problem

This is a **binary classification** task with severe class imbalance (~85% negative). A naive model that predicts "no conversion" for every lead achieves 85% accuracy but provides zero business value. Accuracy is therefore the wrong metric.

The analytic objective is to **rank leads by conversion probability** using AUC-ROC as the primary evaluation metric. A lead scoring model must:

1. Beat a logistic regression benchmark on held-out validation data.
2. Generalize to unseen test data without significant performance degradation (generalization gap < 0.05 AUC).
3. Translate predicted probabilities into a dollar-denominated profit curve that identifies the optimal number of leads to contact.

The deliverable is a scored lead list that lets the sales team allocate effort toward the highest-value opportunities rather than working the pipeline uniformly.

Methodology

The CRISP-DM framework structures the analysis:

- **Data Preparation:** Raw CRM data is loaded, column names standardized, and missing categoricals imputed.
- **Feature Engineering:** Six domain-informed features plus temporal decay, ordinal seniority encoding, and interaction terms.
- **Modeling:** Five gradient boosting models are tuned via randomized search (150 iterations, 5-fold stratified CV), then combined in stacking and voting ensembles.
- **Evaluation:** AUC-ROC is the primary metric. Profit curves translate model scores into dollar-denominated business impact.

Data Preparation & Feature Engineering

```
In [6]: #| label: feature-definitions
#| output: false

# Domain knowledge maps for feature engineering
# Each map encodes business logic from MasterControl's sales process

INTENT_STRENGTH_MAP = {
    'P1 - Website Pricing': 5,
    'P1 - Contact Us': 5,
    'P1 - Video Demo': 3,
    'P1 - Live Demo': 3,
    'P1 - Webinar Demo': 1,
    'No Priority': 1,
    'Priority 1': 2,
    'Priority 2': 0,
}

CHANNEL_TIER_MAP = {
    'Direct/Inbound': 'Premium',
    'SEO': 'Premium',
    'Referrals': 'Premium',
    'Online Ads': 'Standard',
    'Directory Listing': 'Standard',
    'Events': 'Standard',
    'Outbound Prospecting': 'Standard',
    'Email': 'Low-Value',
    'External Demand Gen': 'Low-Value',
}

CHANNEL_NUMERIC_MAP = {
    'Premium': 3, 'Standard': 2,
```

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    'Low-Value': 1, 'Unknown': 2,
}

INDUSTRY_BUDGET_MULTIPLIER = {
    'Pharma & BioTech': 3.0,
    'Blood & Biologics': 2.5,
    'Medical Device': 2.0,
    'Non-Life Science': 1.0,
    'Consumer Packaged Goods': 0.8,
}

TIER_SIZE_MAP = {
    'Small': 50, 'Medium': 500, 'Large': 5000,
}

PRODUCT_ROLE_ALIGNMENT = {
    'Mx': ['Op', 'Mfg', 'Manuf', 'Production', 'Plant'],
    'Qx': ['Qual', 'QA', 'QC', 'Compliance', 'Validation'],
}

HIGH_VALUE_BIGRAMS = [
    'continuous improvement', 'document control',
    'process engineer', 'quality systems',
    'regulatory affairs', 'quality assurance',
    'validation engineer', 'compliance manager',
]

# Ordinal maps for proper rank encoding
SENIORITY_ORDER = [
    'Unknown', 'Other', 'IC', 'Manager',
    'Director', 'VP', 'SVP', 'C-Suite',
]

SCOPE_ORDER = ['Standard', 'Site', 'Regional', 'Global']

```

The raw CRM export contains 16,816 QAL records with account attributes (industry, tier, territory), contact attributes (title, priority), channel source, and pipeline stage. The feature engineering pipeline creates **six domain-informed signals** plus temporal decay, ordinal seniority encoding, and interaction terms.

```

In [7]: #| label: data-pipeline
        #| output: false

# Load raw CRM data, standardize columns, and build all engineered features
def clean_and_engineer(filepath):
    df = pd.read_csv(filepath)
    df.columns = [
        c.strip().lower()
        .replace(' ', '_')
        .replace('/', '_')
        .replace('-', '_')
        for c in df.columns
    ]

    # --- Target ---
    if 'is_success' not in df.columns:

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df['is_success'] = (
    df['next_stage__c']
    .isin(['SQL', 'SQO', 'Won'])
    .astype(int)
)

# --- Feature 1: Intent Strength ---
if 'priority' in df.columns:
    df['intent_strength'] = (
        df['priority']
        .map(INTENT_STRENGTH_MAP).fillna(1)
    )
else:
    df['intent_strength'] = 1

# --- Feature 2: Channel Efficiency ---
ch_col = (
    'last_tactic_campaign_channel'
    if 'last_tactic_campaign_channel' in df.columns
    else 'lead_source'
)
if ch_col in df.columns:
    df['channel_tier'] = (
        df[ch_col].map(CHANNEL_TIER_MAP)
        .fillna('Standard')
    )
    df['channel_efficiency'] = (
        df['channel_tier']
        .map(CHANNEL_NUMERIC_MAP)
    )
else:
    df['channel_tier'] = 'Standard'
    df['channel_efficiency'] = 2

# --- Feature 3: Hidden Gem Flag ---
m_col = (
    'acct_manufacturing_model'
    if 'acct_manufacturing_model' in df.columns
    else None
)
i_col = (
    'acct_target_industry'
    if 'acct_target_industry' in df.columns
    else None
)
s_col = (
    'acct_primary_site_function'
    if 'acct_primary_site_function' in df.columns
    else None
)
gem = pd.Series(False, index=df.index)
if m_col:
    gem |= df[m_col].str.contains(
        'Not Enough Info', case=False, na=False
    )
if s_col:

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    gem |= df[s_col].str.contains(
        'Non-manufacturing', case=False, na=False
    )
if i_col:
    gem |= df[i_col].str.contains(
        'Non-manufacturing', case=False, na=False
    )
df['is_hidden_gem'] = gem.astype(int)

# --- Feature 4: Capital Density Score ---
t_col = (
    'acct_tier_rollup'
    if 'acct_tier_rollup' in df.columns
    else None
)
if i_col and t_col:
    df['_ind_mult'] = df[i_col].map(
        lambda x: next(
            (v for k, v
             in INDUSTRY_BUDGET_MULTIPLIER.items()
             if k.lower() in str(x).lower()),
            1.0,
        )
    )
    df['_tier_size'] = (
        df[t_col].map(TIER_SIZE_MAP).fillna(500)
    )
    df['capital_density_score'] = (
        df['_ind_mult'] * df['_tier_size']
    )
    df['capital_density_log'] = np.log1p(
        df['capital_density_score']
    )
    df.drop(
        columns=['_ind_mult', '_tier_size',
                 'capital_density_score'],
        errors='ignore', inplace=True,
    )
else:
    df['capital_density_log'] = np.log1p(500)

# --- Feature 5: Role-Product Match (continuous) ---
title_col = (
    'contact_lead_title'
    if 'contact_lead_title' in df.columns
    else None
)
p_col = (
    'product_segment'
    if 'product_segment' in df.columns
    else 'solution_rollup'
)
if 'product_segment' not in df.columns:
    df['product_segment'] = df['solution_rollup'].apply(
        lambda s: (
            'Mx' if str(s) == 'Mx'

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        else ('Qx' if str(s) == 'Qx'
              else 'Other')
    )
)
if title_col and p_col in df.columns:
    def _role_match(row):
        title = (
            str(row[title_col]).lower()
            if pd.notna(row[title_col]) else ''
        )
        product = (
            str(row[p_col])
            if pd.notna(row[p_col]) else ''
        )
        if product in PRODUCT_ROLE_ALIGNMENT:
            hits = sum(
                1 for kw
                in PRODUCT_ROLE_ALIGNMENT[product]
                if kw.lower() in title
            )
            return hits
        return 0
    df['role_product_match'] = df.apply(
        _role_match, axis=1
    )
else:
    df['role_product_match'] = 0

# --- Feature 6: Title Bigrams ---
if title_col and title_col in df.columns:
    for bigram in HIGH_VALUE_BIGRAMS:
        col_name = 'has_' + bigram.replace(' ', '_')
        df[col_name] = (
            df[title_col].str.lower()
            .str.contains(bigram, na=False)
            .astype(int)
        )
    bg_cols = [
        c for c in df.columns if c.startswith('has_')
    ]
    df['title_bigram_count'] = df[bg_cols].sum(axis=1)
else:
    df['title_bigram_count'] = 0

# --- Title Parsing ---
if (
    'title_seniority' not in df.columns
    and title_col
    and title_col in df.columns
):
    def _seniority(t):
        if pd.isna(t):
            return 'Unknown'
        t = str(t).lower()
        if re.search(
            r'\b(ceo|cfo|coo|cto|cio|chief|'

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        r'c-level|president)\b', t
    ):
        return 'C-Suite'
    if re.search(
        r'\b(svp|senior vice president|evp)\b', t
    ):
        return 'SVP'
    if re.search(
        r'\b(vp|vice president)\b', t
    ):
        return 'VP'
    if re.search(
        r'\b(director|head of)\b', t
    ):
        return 'Director'
    if re.search(
        r'\b(manager|mgr|supervisor|lead)\b', t
    ):
        return 'Manager'
    if re.search(
        r'\b(analyst|engineer|specialist|'
        r'associate|coordinator)\b', t
    ):
        return 'IC'
    return 'Other'

def _function(t):
    if pd.isna(t):
        return 'Unknown'
    t = str(t).lower()
    if re.search(
        r'\b(quality|qa|qc|qms|compliance|'
        r'validation|capa)\b', t
    ):
        return 'Quality'
    if re.search(
        r'\b(regulatory|reg affairs|'
        r'submissions)\b', t
    ):
        return 'Regulatory'
    if re.search(
        r'\b(manufacturing|production|'
        r'operations|ops|plant|supply)\b', t
    ):
        return 'Mfg/Ops'
    if re.search(
        r'\b(it|information tech|software|'
        r'systems|data)\b', t
    ):
        return 'IT'
    if re.search(
        r'\b(r&d|research|development|'
        r'scientist|clinical|lab)\b', t
    ):
        return 'R&D'
    if re.search(

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        r'\b(project|program|pmo)\b', t
    ):
        return 'PMO'
    return 'Other'

def _scope(t):
    if pd.isna(t):
        return 'Standard'
    t = str(t).lower()
    if re.search(
        r'\b(global|worldwide|international|
        r'corporate|enterprise)\b', t
    ):
        return 'Global'
    if re.search(
        r'\b(regional|division|group)\b', t
    ):
        return 'Regional'
    if re.search(
        r'\b(site|plant|facility|local)\b', t
    ):
        return 'Site'
    return 'Standard'

df['title_seniority'] = (
    df[title_col].apply(_seniority)
)
df['title_function'] = (
    df[title_col].apply(_function)
)
df['title_scope'] = (
    df[title_col].apply(_scope)
)

# --- NEW: Ordinal seniority & scope encoding ---
if 'title_seniority' in df.columns:
    sen_map = {
        v: i for i, v in enumerate(SENIORITY_ORDER)
    }
    df['seniority_rank'] = (
        df['title_seniority'].map(sen_map).fillna(0)
    )
if 'title_scope' in df.columns:
    sco_map = {
        v: i for i, v in enumerate(SCOPE_ORDER)
    }
    df['scope_rank'] = (
        df['title_scope'].map(sco_map).fillna(0)
    )

# --- Decision maker ---
if 'is_decision_maker' not in df.columns:
    df['is_decision_maker'] = (
        df['title_seniority']
        .isin(['C-Suite', 'SVP', 'VP', 'Director'])
        .astype(int)

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    )

# --- Title specificity (word count) ---
if title_col and title_col in df.columns:
    df['title_word_count'] = (
        df[title_col].fillna('')
        .apply(lambda x: len(str(x).split()))
    )

# --- Temporal features ---
if (
    'cohort_date' in df.columns
    or 'qal_cohort_date' in df.columns
):
    cohort_src = (
        'qal_cohort_date'
        if 'qal_cohort_date' in df.columns
        else 'cohort_date'
    )
    df['cohort_date'] = pd.to_datetime(
        df[cohort_src], errors='coerce'
    )
    if 'lead_age_days' not in df.columns:
        df['lead_age_days'] = (
            df['cohort_date'].max()
            - df['cohort_date']
        ).dt.days

    if 'lead_age_days' in df.columns:
        df['velocity_tier'] = pd.cut(
            df['lead_age_days'].fillna(0),
            bins=[-1, 30, 60, 90, 180, 9999],
            labels=['Hot', 'Warm', 'Cooling',
                    'Cold', 'Stale'],
        ).astype(str)
        df['is_fresh'] = (
            (df['lead_age_days'] <= 30).astype(int)
        )
        df['is_stale'] = (
            (df['lead_age_days'] > 180).astype(int)
        )
        # NEW: exponential decay
        df['lead_age_decay'] = np.exp(
            -df['lead_age_days'].fillna(0) / 90
        )

# --- Interaction features ---
sen_c = (
    'title_seniority'
    if 'title_seniority' in df.columns else None
)
ind_c = (
    'acct_target_industry'
    if 'acct_target_industry' in df.columns
    else None
)

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mod_c = (
    'acct_manufacturing_model'
    if 'acct_manufacturing_model' in df.columns
    else None
)
if sen_c and ind_c and mod_c:
    df['seniority_x_industry'] = (
        df[sen_c].astype(str) + '_'
        + df[ind_c].astype(str)
    )
    df['seniority_x_model'] = (
        df[sen_c].astype(str) + '_'
        + df[mod_c].astype(str)
    )
    df['industry_x_model'] = (
        df[ind_c].astype(str) + '_'
        + df[mod_c].astype(str)
    )
    df['power_trio'] = (
        df[sen_c].astype(str) + '_'
        + df[ind_c].astype(str) + '_'
        + df[mod_c].astype(str)
    )
    senior = df[sen_c].isin(
        ['Director', 'VP', 'SVP', 'C-Suite']
    )
    pharma = df[ind_c].str.contains(
        'Pharma|Life|Bio', case=False, na=False
    )
    inhouse = df[mod_c].str.contains(
        'In-House|In House|Inhouse',
        case=False, na=False,
    )
    df['is_golden_segment'] = (
        (senior & pharma & inhouse).astype(int)
    )
    df['is_senior_pharma'] = (
        (senior & pharma).astype(int)
    )

if 'title_scope' in df.columns:
    df['is_global_scope'] = (
        (df['title_scope'] == 'Global').astype(int)
    )

# NEW: industry-channel synergy
if ind_c and 'channel_tier' in df.columns:
    df['industry_channel_synergy'] = (
        (
            df[ind_c].str.contains(
                'Pharma|Bio', case=False, na=False
            )
            & (df['channel_tier'] == 'Premium')
        ).astype(int)
    )

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# NEW: seniority × decision-relevance weighting
if 'title_function' in df.columns:
    decision_funcs = ['Quality', 'Regulatory']
    df['is_decision_relevant'] = (
        (
            df['is_decision_maker'].eq(1)
            & df['title_function'].isin(decision_funcs)
        ).astype(int)
    )

# Fill missing categoricals
for col in [
    'acct_manufacturing_model',
    'acct_primary_site_function',
    'acct_target_industry',
    'acct_territory_rollup',
    'title_seniority', 'title_function',
    'title_scope', 'channel_tier',
]:
    if col in df.columns:
        df[col] = df[col].fillna('Unknown')

return df

df = clean_and_engineer(DATA_PATH)

```

```

In [8]: #| label: dataset-overview

n_leads = len(df)
target_rate = df['is_success'].mean()
n_mx = (df['product_segment'] == 'Mx').sum()
n_qx = (df['product_segment'] == 'Qx').sum()
mx_rate = df.loc[
    df['product_segment'] == 'Mx', 'is_success'
].mean()
qx_rate = df.loc[
    df['product_segment'] == 'Qx', 'is_success'
].mean()
n_dm = df['is_decision_maker'].sum()

summary_data = pd.DataFrame({
    'Metric': [
        'Total Leads', 'Target Rate (SQL+)',
        'Mx Leads', 'Qx Leads',
        'Decision Makers', 'Features Engineered',
    ],
    'Value': [
        f'{n_leads:,}',
        f'{target_rate:.1%}',
        f'{n_mx:,} ({mx_rate:.1%} conv.)',
        f'{n_qx:,} ({qx_rate:.1%} conv.)',
        f'{n_dm:,} ({n_dm/n_leads:.1%})',
        '6 core + ordinal + decay + interactions',
    ],
})

```



```
})
print_table(summary_data, caption="Dataset Overview")
```

****Dataset Overview****

Metric	Value
Total Leads	16,815
Target Rate (SQL+)	17.9%
Mx Leads	4,239 (12.6% conv.)
Qx Leads	12,547 (19.7% conv.)
Decision Makers	3,066 (18.2%)
Features Engineered	6 core + ordinal + decay + interactions

Understanding the Data at a Glance

Target Distribution & Product Segment Gap

```
In [9]: #| label: target-and-product-gap

fig, axes = plt.subplots(1, 3, figsize=(7.5, 3.5))

counts = df['is_success'].value_counts().sort_index()
labels_pie = ['Not Converted', 'Converted (SQL+)']
colors_t = [COL_NEUTRAL, COL_SUCCESS]
wedges, texts, autotexts = axes[0].pie(
    counts, labels=labels_pie, colors=colors_t,
    autopct='%1.1f%%', startangle=90,
    textprops={'fontsize': 8},
)
for at in autotexts:
    at.set_fontweight('bold')
axes[0].set_title(
    'Target Distribution',
    fontweight='bold', fontsize=10,
)

seg_conv = (
    df.groupby('product_segment')['is_success']
    .agg(['mean', 'count'])
    .reindex(['Mx', 'Qx', 'Other']).dropna()
)
bar_colors = [
    COL_RISK, COL_SUCCESS, COL_NEUTRAL
][:len(seg_conv)]
axes[1].bar(
    seg_conv.index, seg_conv['mean'],
    color=bar_colors, edgecolor='white', linewidth=1,
)
for i, (idx, row) in enumerate(seg_conv.iterrows()):
    axes[1].text(
        i, row['mean'] + 0.005,
        f'{row["mean"]:.1%}',
        ha='center', fontweight='bold', fontsize=9,
```

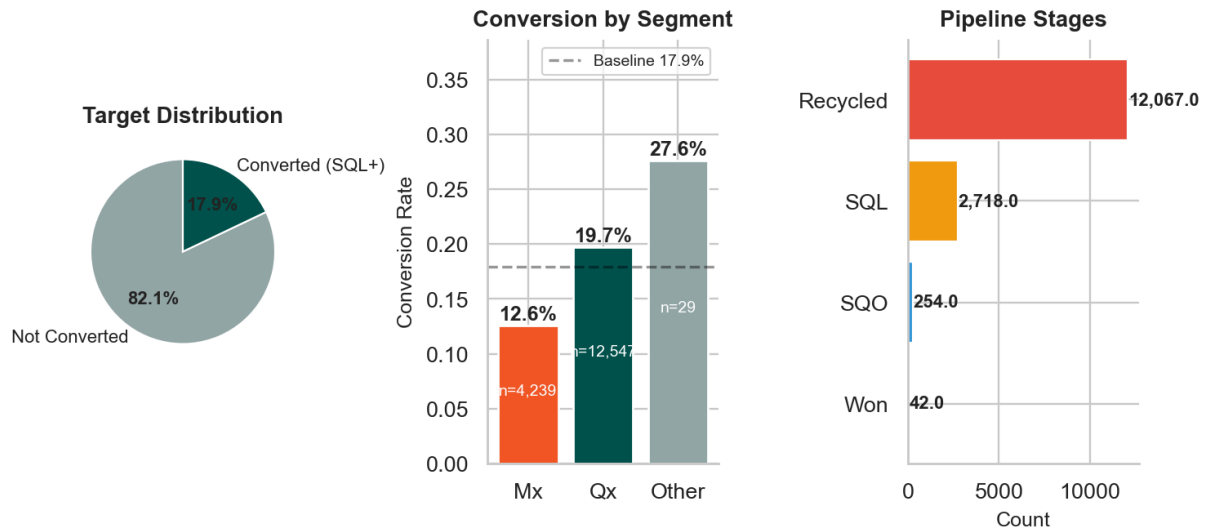
```

    )
    axes[1].text(
        i, row['mean'] / 2,
        f'n={int(row["count"]):,}',
        ha='center', color='white', fontsize=7,
    )
    axes[1].axhline(
        y=target_rate, color='black',
        linestyle='--', alpha=0.4,
        label=f'Baseline {target_rate:.1%}',
    )
    axes[1].set_ylabel('Conversion Rate', fontsize=9)
    axes[1].set_title(
        'Conversion by Segment',
        fontweight='bold', fontsize=10,
    )
    axes[1].legend(fontsize=7)
    axes[1].set_ylim(0, seg_conv['mean'].max() * 1.4)

    if 'next_stage__c' in df.columns:
        stage_order = [
            'Disqualified', 'Recycled',
            'SQL', 'SQ0', 'Won',
        ]
        stage_counts = (
            df['next_stage__c'].value_counts()
            .reindex(stage_order).dropna()
        )
        stage_colors = [
            '#e74c3c', '#f39c12', '#3498db',
            '#2ecc71', '#00534B',
        ][0:len(stage_counts)]
        axes[2].barh(
            stage_counts.index, stage_counts.values,
            color=stage_colors, edgecolor='white',
        )
        for i, (idx, v) in enumerate(
            stage_counts.items()
        ):
            axes[2].text(
                v + 50, i, f'{v:,}',
                va='center', fontweight='bold', fontsize=8,
            )
        axes[2].set_xlabel('Count', fontsize=9)
        axes[2].set_title(
            'Pipeline Stages',
            fontweight='bold', fontsize=10,
        )
        axes[2].invert_yaxis()

    for ax in axes:
        sns.despine(ax=ax)
    plt.tight_layout()
    plt.show()

```



The **class imbalance** is immediately visible -only ~15% of leads convert. The 7 percentage-point gap between Mx and Qx segments confirms the business challenge.

Channel Tier Conversion Rates

```
In [10]: #| label: channel-tier-conversion

fig, axes = plt.subplots(1, 2, figsize=(7.5, 3.5))

channel_conv = (
    df.groupby('channel_tier')['is_success']
      .agg(['mean', 'count'])
      .reindex(['Premium', 'Standard', 'Low-Value'])
)

colors_ch = [COL_PREMIUM, COL_NEUTRAL, COL_LOWVAL]
axes[0].bar(
    channel_conv.index, channel_conv['mean'],
    color=colors_ch, edgecolor='white', linewidth=1,
)

for i, (idx, row) in enumerate(channel_conv.iterrows()):
    axes[0].text(
        i, row['mean'] + 0.008,
        f'{row["mean"]:.1%}',
        ha='center', fontweight='bold', fontsize=10,
    )
    axes[0].text(
        i, row['mean'] / 2,
        f'n={int(row["count"]):,}',
        ha='center', color='white', fontsize=7,
    )

axes[0].axhline(
    y=target_rate, color='black',
    linestyle='--', alpha=0.4,
    label=f'Baseline ({target_rate:.1%})',
)

axes[0].set_ylabel('Conversion Rate', fontsize=9)
axes[0].set_title(
```

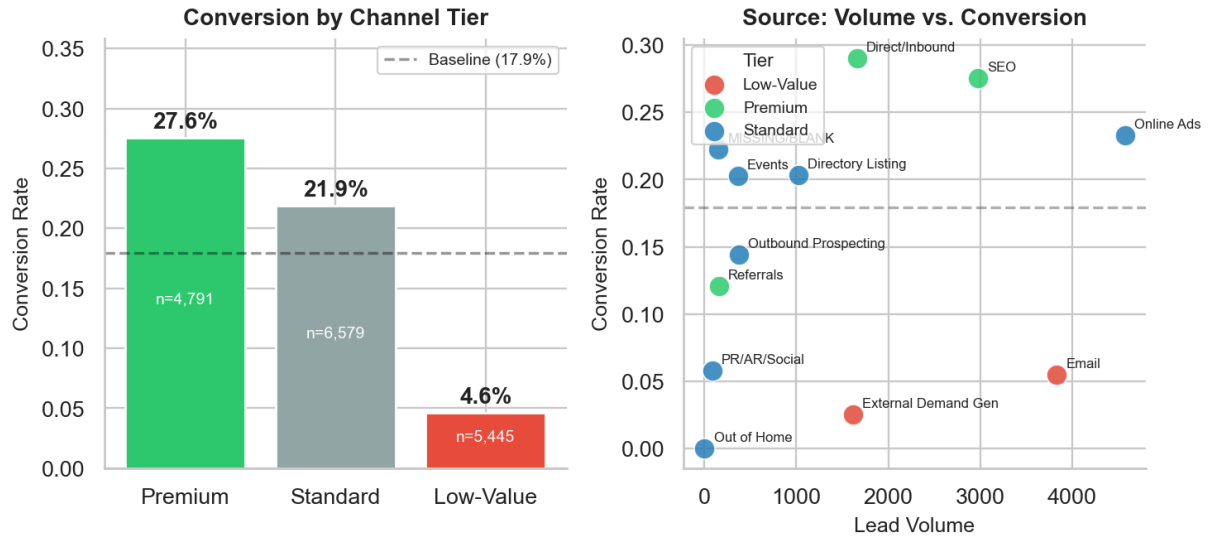
```

        'Conversion by Channel Tier',
        fontweight='bold', fontsize=10,
    )
    axes[0].set_ylim(0, channel_conv['mean'].max() * 1.3)
    axes[0].legend(fontsize=7)
    sns.despine(ax=axes[0])

    src_col = (
        'last_tactic_campaign_channel'
        if 'last_tactic_campaign_channel' in df.columns
        else 'lead_source'
    )
    if src_col in df.columns:
        src_stats = (
            df.groupby(src_col)['is_success']
            .agg(['mean', 'count']).reset_index()
        )
        src_stats.columns = ['Source', 'ConvRate', 'Volume']
        src_stats['Tier'] = (
            src_stats['Source']
            .map(CHANNEL_TIER_MAP).fillna('Standard')
        )
        tier_colors = {
            'Premium': COL_PREMIUM,
            'Standard': COL_ACCENT,
            'Low-Value': COL_LOWVAL,
        }
        for tier, grp in src_stats.groupby('Tier'):
            axes[1].scatter(
                grp['Volume'], grp['ConvRate'], s=80,
                color=tier_colors.get(tier, COL_NEUTRAL),
                label=tier, edgecolor='white',
                linewidth=0.6, alpha=0.85, zorder=3,
            )
        for _, row in src_stats.iterrows():
            axes[1].annotate(
                row['Source'],
                (row['Volume'], row['ConvRate']),
                fontsize=6, ha='left', va='bottom',
                xytext=(4, 2), textcoords='offset points',
            )
        axes[1].axhline(
            y=target_rate, color='black',
            linestyle='--', alpha=0.3,
        )
        axes[1].set_xlabel('Lead Volume', fontsize=9)
        axes[1].set_ylabel('Conversion Rate', fontsize=9)
        axes[1].set_title(
            'Source: Volume vs. Conversion',
            fontweight='bold', fontsize=10,
        )
        axes[1].legend(
            title='Tier', fontsize=7, title_fontsize=8,
        )
    sns.despine(ax=axes[1])

```

```
plt.tight_layout()
plt.show()
```



Low-value channels convert at roughly **one-fifth** the rate of premium channels, yet account for significant pipeline volume.

Seniority & Function Heatmap

```
In [11]: #| label: seniority-function-heatmap

fig, axes = plt.subplots(1, 2, figsize=(7.5, 3.8))

if 'title_seniority' in df.columns:
    sen_order = [
        'C-Suite', 'SVP', 'VP', 'Director',
        'Manager', 'IC', 'Other', 'Unknown',
    ]
    sen_conv = (
        df.groupby('title_seniority')['is_success']
        .agg(['mean', 'count'])
    )
    sen_conv = sen_conv.reindex(
        [s for s in sen_order if s in sen_conv.index]
    )
    bar_cols = [
        COL_SUCCESS if v > target_rate else COL_NEUTRAL
        for v in sen_conv['mean']
    ]
    axes[0].barh(
        sen_conv.index, sen_conv['mean'],
        color=bar_cols, edgecolor='white',
    )
    for i, (idx, row) in enumerate(
        sen_conv.iterrows()
    ):
        axes[0].text(
            row['mean'] + 0.003, i,
```

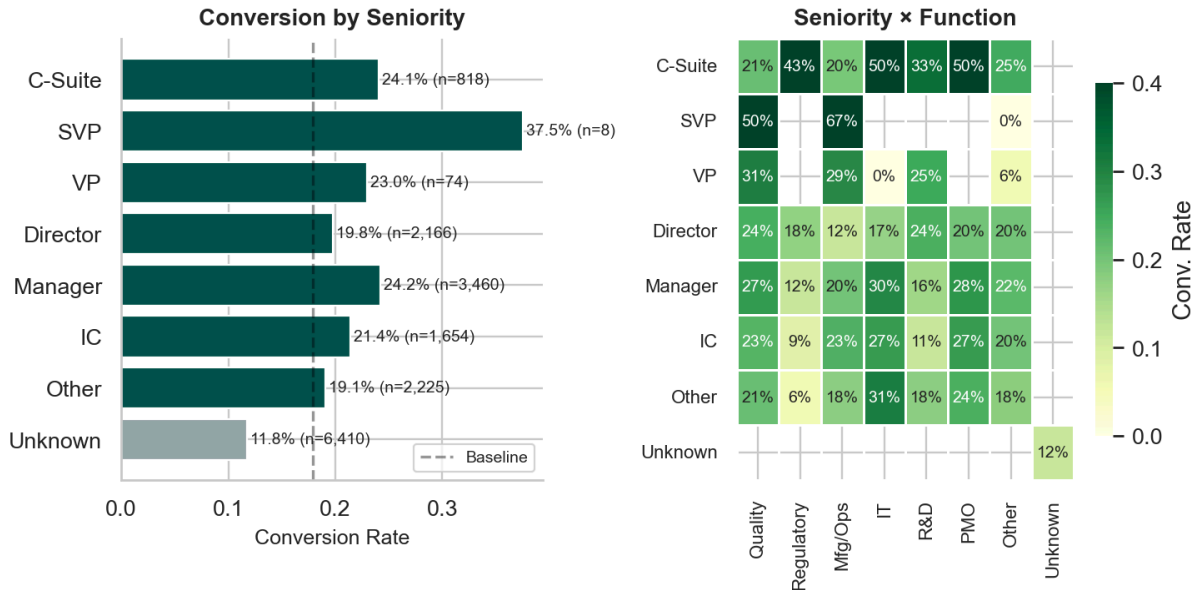
```

        f'{row["mean"]:.1%}'
        f'(n={int(row["count"]):,})',
        va='center', fontsize=7,
    )
    axes[0].axvline(
        x=target_rate, color='black',
        linestyle='--', alpha=0.4, label='Baseline',
    )
    axes[0].set_xlabel('Conversion Rate', fontsize=9)
    axes[0].set_title(
        'Conversion by Seniority',
        fontweight='bold', fontsize=10,
    )
    axes[0].invert_yaxis()
    axes[0].legend(fontsize=7)
    sns.despine(ax=axes[0])

if (
    'title_seniority' in df.columns
    and 'title_function' in df.columns
):
    pivot = df.pivot_table(
        values='is_success',
        index='title_seniority',
        columns='title_function', aggfunc='mean',
    )
    sen_re = [
        s for s in sen_order if s in pivot.index
    ]
    func_order = [
        'Quality', 'Regulatory', 'Mfg/Ops',
        'IT', 'R&D', 'PMO', 'Other', 'Unknown',
    ]
    func_re = [
        f for f in func_order if f in pivot.columns
    ]
    pivot = pivot.reindex(
        index=sen_re, columns=func_re
    )
    sns.heatmap(
        pivot, annot=True, fmt='.0%', cmap='YlGn',
        ax=axes[1], linewidths=0.5,
        cbar_kws={
            'label': 'Conv. Rate', 'shrink': 0.8,
        },
        vmin=0, vmax=0.4, annot_kws={'size': 7},
    )
    axes[1].set_title(
        'Seniority × Function',
        fontweight='bold', fontsize=10,
    )
    axes[1].set_ylabel('')
    axes[1].set_xlabel('')
    axes[1].tick_params(labelsize=8)

```

```
plt.tight_layout()
plt.show()
```



VPs and Directors in Quality and Regulatory convert at the highest rates -validating the role-product match feature.

Engineered Feature Distributions

```
In [12]: #| label: feature-distributions

fig, axes = plt.subplots(2, 3, figsize=(7.5, 5))

eng_features = [
    ('intent_strength', 'Intent Strength', True),
    ('channel_efficiency', 'Channel Efficiency', True),
    ('capital_density_log', 'Capital Density (log)', False),
    ('is_hidden_gem', 'Hidden Gem Flag', True),
    ('role_product_match', 'Role-Product Match', True),
    ('lead_age_decay', 'Lead Age Decay', False),
]

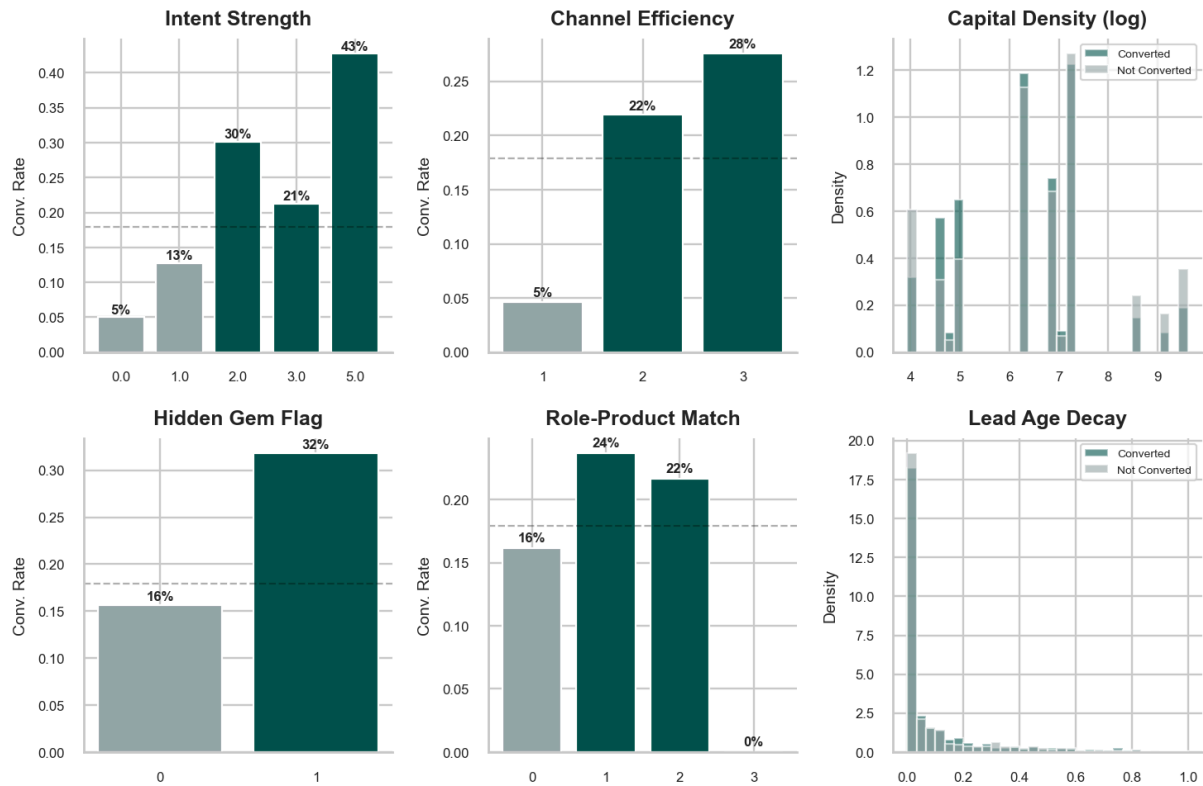
for ax, (feat, feat_label, is_discrete) in zip(
    axes.flat, eng_features
):
    if feat not in df.columns:
        ax.set_visible(False)
        continue
    if is_discrete:
        vals = sorted(df[feat].dropna().unique())
        conv = df.groupby(feat)['is_success'].mean()
        bc = [
            COL_SUCCESS if conv.get(v, 0) > target_rate
            else COL_NEUTRAL
            for v in vals
        ]
        ax.bar(
```

```

        [str(v) for v in vals],
        [conv.get(v, 0) for v in vals],
        color=bc, edgecolor='white',
    )
    for i, v in enumerate(vals):
        ax.text(
            i, conv.get(v, 0) + 0.005,
            f'{conv.get(v, 0):.0%}',
            ha='center', fontsize=6,
            fontweight='bold',
        )
    ax.axhline(
        y=target_rate, color='black',
        linestyle='--', alpha=0.3, linewidth=0.8,
    )
    ax.set_ylabel('Conv. Rate', fontsize=7)
else:
    for lv, color, lbl in [
        (1, COL_SUCCESS, 'Converted'),
        (0, COL_NEUTRAL, 'Not Converted'),
    ]:
        subset = df[df['is_success'] == lv][feat].dropna()
        ax.hist(
            subset, bins=30, alpha=0.6,
            color=color, label=lbl, density=True,
        )
    ax.legend(fontsize=5)
    ax.set_ylabel('Density', fontsize=7)
    ax.set_title(feat_label, fontweight='bold', fontsize=9)
    ax.tick_params(labelsize=6)
    sns.despine(ax=ax)

plt.tight_layout()
plt.show()

```

Each feature shows separation between converted and non-converted leads. The new **lead age decay** feature (bottom-right) captures smooth exponential decline rather than crude velocity bins.

Feature Correlation Matrix

```
In [13]: #| label: feature-correlation

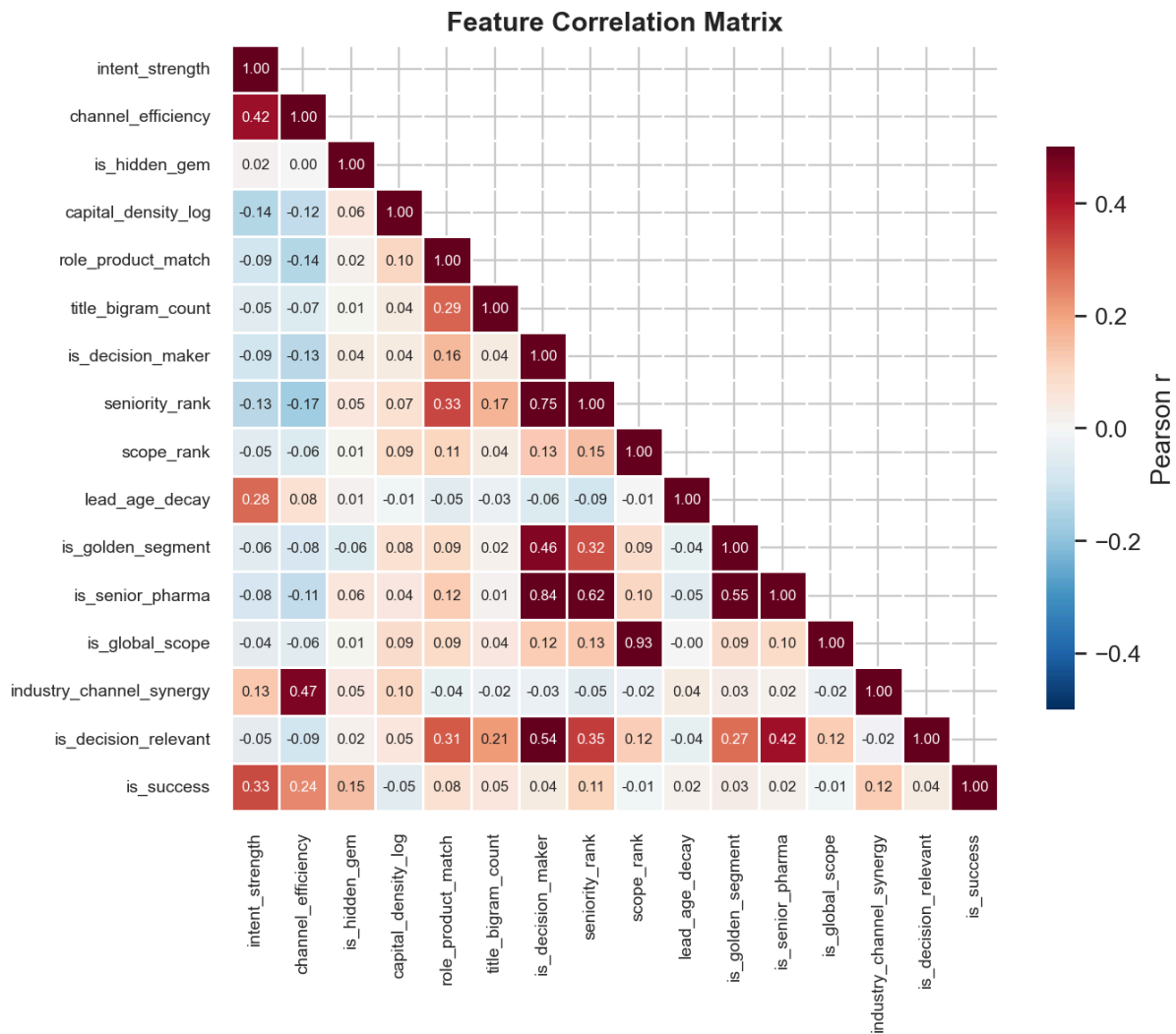
corr_features = [c for c in [
    'intent_strength', 'channel_efficiency',
    'is_hidden_gem', 'capital_density_log',
    'role_product_match', 'title_bigram_count',
    'is_decision_maker', 'seniority_rank',
    'scope_rank', 'lead_age_decay',
    'is_golden_segment', 'is_senior_pharma',
    'is_global_scope', 'industry_channel_synergy',
    'is_decision_relevant', 'is_success',
]] if c in df.columns

fig, ax = plt.subplots(figsize=(7, 6))
corr_matrix = df[corr_features].corr()
mask = np.triu(
    np.ones_like(corr_matrix, dtype=bool), k=1
)
sns.heatmap(
    corr_matrix, mask=mask, annot=True, fmt='.2f',
    cmap='RdBu_r', center=0, vmin=-0.5, vmax=0.5,
    ax=ax, linewidths=0.5,
    cbar_kws={'shrink': 0.7, 'label': 'Pearson r'},
    square=True, annot_kws={'size': 6},
```

```

)
ax.set_title(
    'Feature Correlation Matrix',
    fontweight='bold', fontsize=11,
)
ax.tick_params(labelsize=7)
plt.tight_layout()
plt.show()

```



Low inter-feature correlations confirm the features capture **distinct signals** -ideal for ensemble modeling.

Model Analysis

```

In [14]: #| label: feature-matrix
          #| output: false

# Separate features into categorical, numeric, and text for the preprocessing
def prepare_feature_matrix(df):
    y = df['is_success'].values
    categorical_features = [c for c in [

```

```

        'title_seniority', 'title_function',
        'title_scope', 'acct_target_industry',
        'acct_manufacturing_model',
        'acct_primary_site_function',
        'acct_territory_rollup',
        'product_segment', 'channel_tier',
    ] if c in df.columns]

    interaction_features = [c for c in [
        'seniority_x_industry', 'seniority_x_model',
        'industry_x_model', 'power_trio',
    ] if c in df.columns]

    velocity_cats = [
        c for c in ['velocity_tier']
        if c in df.columns
    ]
    all_categoricals = (
        categorical_features
        + interaction_features
        + velocity_cats
    )

    numeric_features = [c for c in [
        'lead_age_days', 'is_decision_maker',
        'is_fresh', 'is_stale',
        'is_golden_segment', 'is_senior_pharma',
        'is_global_scope', 'intent_strength',
        'channel_efficiency', 'is_hidden_gem',
        'capital_density_log', 'role_product_match',
        'title_bigram_count',
        'seniority_rank', 'scope_rank',
        'lead_age_decay', 'title_word_count',
        'industry_channel_synergy',
        'is_decision_relevant',
    ] if c in df.columns]

    bg_cols = [
        c for c in df.columns if c.startswith('has_')
    ]
    numeric_features.extend(bg_cols)

    text_col = (
        'contact_lead_title'
        if 'contact_lead_title' in df.columns
        else None
    )
    X = df[all_categoricals + numeric_features].copy()
    text_data = (
        df[text_col].fillna('')
        if text_col else None
    )
    return (
        X, y, text_data,
        all_categoricals, numeric_features,
    )

```

```
X, y, text_data, cat_cols, num_cols = (
    prepare_feature_matrix(df)
)
```

```
In [15]: #| label: data-split
#| output: false

# 80/10/10 stratified split: train, validation (model selection), test (final)
X_temp, X_test, y_temp, y_test = train_test_split(
    X, y, test_size=TEST_SIZE,
    random_state=RANDOM_STATE, stratify=y,
)
X_train, X_val, y_train, y_val = train_test_split(
    X_temp, y_temp,
    test_size=VAL_SIZE / (1 - TEST_SIZE),
    random_state=RANDOM_STATE, stratify=y_temp,
)

if text_data is not None:
    text_temp, text_test = train_test_split(
        text_data, test_size=TEST_SIZE,
        random_state=RANDOM_STATE, stratify=y,
    )
    text_train, text_val = train_test_split(
        text_temp,
        test_size=VAL_SIZE / (1 - TEST_SIZE),
        random_state=RANDOM_STATE, stratify=y_temp,
    )
else:
    text_train = text_val = text_test = None
```

```
In [16]: #| label: encoding
#| output: false

# Target encoding for high-cardinality categoricals (>5 levels)
# Label encoding for low-cardinality, TF-IDF + LSA for title text
target_encode_cols = [
    c for c in cat_cols if X_train[c].nunique() > 5
]
standard_encode_cols = [
    c for c in cat_cols
    if c not in target_encode_cols
]

class ManualTargetEncoder(BaseEstimator, TransformerMixin):
    def __init__(self, columns=None, smoothing=10):
        self.columns = columns
        self.smoothing = smoothing
        self.encoding_maps_ = {}
        self.global_mean_ = None

    def fit(self, X, y):
        X = (
            pd.DataFrame(X)
```

```

        if not isinstance(X, pd.DataFrame)
        else X
    )
    y = np.array(y)
    self.global_mean_ = y.mean()
    cols = (
        self.columns
        or X.select_dtypes(
            include=['object', 'category']
        ).columns.tolist()
    )
    for col in cols:
        if col in X.columns:
            tmp = pd.DataFrame({
                'col': X[col].astype(str),
                'target': y,
            })
            agg = tmp.groupby('col')['target'].agg(
                ['mean', 'count']
            )
            smoothed = (
                (agg['count'] * agg['mean']
                 + self.smoothing
                 * self.global_mean_)
                / (agg['count'] + self.smoothing)
            )
            self.encoding_maps_[col] = (
                smoothed.to_dict()
            )
    return self

def transform(self, X):
    X = (
        pd.DataFrame(X).copy()
        if not isinstance(X, pd.DataFrame)
        else X.copy()
    )
    for col, mapping in self.encoding_maps_.items():
        if col in X.columns:
            X[col + '_encoded'] = (
                X[col].astype(str)
                .map(mapping)
                .fillna(self.global_mean_)
            )
    return X

if TARGET_ENCODER_AVAILABLE and len(target_encode_cols) > 0:
    target_encoder = TargetEncoder(
        smooth='auto', target_type='binary',
    )
    X_train_te = X_train.copy()
    X_val_te = X_val.copy()
    X_test_te = X_test.copy()
    te_train = target_encoder.fit_transform(
        X_train[target_encode_cols], y_train,
    )

```

```

te_val = target_encoder.transform(
    X_val[target_encode_cols],
)
te_test = target_encoder.transform(
    X_test[target_encode_cols],
)
for i, col in enumerate(target_encode_cols):
    X_train_te[col] = te_train[:, i]
    X_val_te[col] = te_val[:, i]
    X_test_te[col] = te_test[:, i]
elif len(target_encode_cols) > 0:
    manual_enc = ManualTargetEncoder(
        columns=target_encode_cols, smoothing=10,
    )
    X_train_te = manual_enc.fit_transform(
        X_train, y_train,
    )
    X_val_te = manual_enc.transform(X_val)
    X_test_te = manual_enc.transform(X_test)
    for col in target_encode_cols:
        enc_col = col + '_encoded'
        if enc_col in X_train_te.columns:
            for d in [X_train_te, X_val_te, X_test_te]:
                d[col] = d[enc_col]
else:
    X_train_te = X_train.copy()
    X_val_te = X_val.copy()
    X_test_te = X_test.copy()

label_encoders = {}
for col in standard_encode_cols:
    le = LabelEncoder()
    X_train_te[col] = le.fit_transform(
        X_train_te[col].astype(str)
    )
    safe_tx = lambda s, enc=le: s.astype(str).apply(
        lambda x: (
            enc.transform([x])[0]
            if x in enc.classes_ else 0
        )
    )
    X_val_te[col] = safe_tx(X_val_te[col])
    X_test_te[col] = safe_tx(X_test_te[col])
    label_encoders[col] = le

if text_train is not None:
    tfidf = TfidfVectorizer(
        max_features=TFIDF_MAX_FEATURES,
        ngram_range=(1, 2),
        stop_words='english', min_df=5,
    )
    svd = TruncatedSVD(
        n_components=LSA_COMPONENTS,
        random_state=RANDOM_STATE,
    )
    lsa_train = svd.fit_transform(

```

```

        tfidf.fit_transform(text_train)
    )
    lsa_val = svd.transform(
        tfidf.transform(text_val)
    )
    lsa_test = svd.transform(
        tfidf.transform(text_test)
    )
    for i in range(LSA_COMPONENTS):
        col = f'lsa_{i}'
        X_train_te[col] = lsa_train[:, i]
        X_val_te[col] = lsa_val[:, i]
        X_test_te[col] = lsa_test[:, i]

for col in X_train_te.columns:
    if X_train_te[col].dtype == 'object':
        le = LabelEncoder()
        X_train_te[col] = le.fit_transform(
            X_train_te[col].astype(str)
        )
        safe_enc = lambda s, enc=le: s.astype(str).apply(
            lambda x: (
                enc.transform([x])[0]
                if x in enc.classes_ else 0
            )
        )
        X_val_te[col] = safe_enc(X_val_te[col])
        X_test_te[col] = safe_enc(X_test_te[col])

X_train_te = X_train_te.fillna(0)
X_val_te = X_val_te.fillna(0)
X_test_te = X_test_te.fillna(0)

# SMOTE is applied inside CV folds via imblearn Pipeline (see model training
# This avoids synthetic-sample leakage across CV folds
if SMOTE_AVAILABLE:
    from imblearn.pipeline import Pipeline as ImbPipeline

```

In [17]: `#| label: split-summary`

```

split_summary = pd.DataFrame({
    'Partition': ['Train', 'Validation', 'Test'],
    'Rows': [
        f'{len(X_train):,}',
        f'{len(X_val):,}',
        f'{len(X_test):,}',
    ],
    'Positive Rate': [
        f'{y_train.mean():.1%}',
        f'{y_val.mean():.1%}',
        f'{y_test.mean():.1%}',
    ],
    'Features': [X_train_te.shape[1]] * 3,
})
print_table(
    split_summary,

```

```
caption="Data Partition Summary (80/10/10, SMOTE applied inside CV folds
)
```

```
**Data Partition Summary (80/10/10, SMOTE applied inside CV folds)**
```

Partition	Rows	Positive Rate	Features
Train	13,451	17.9%	61
Validation	1,682	18.0%	61
Test	1,682	17.9%	61

Benchmark Model

A performance benchmark establishes the floor. We use logistic regression with balanced class weights - a simple, interpretable model that any candidate must beat to justify added complexity.

```
In [18]: #| label: benchmark-model

# Logistic regression benchmark - minimum bar for candidate models
benchmark_lr = LogisticRegression(
    random_state=RANDOM_STATE,
    max_iter=1000, class_weight='balanced',
)
benchmark_lr.fit(X_train_te, y_train)

# In-sample (train) and out-of-sample (validation) AUC
bench_train_probs = benchmark_lr.predict_proba(X_train_te)[: , 1]
bench_val_probs = benchmark_lr.predict_proba(X_val_te)[: , 1]
bench_train_auc = roc_auc_score(y_train, bench_train_probs)
bench_val_auc = roc_auc_score(y_val, bench_val_probs)

bench_df = pd.DataFrame({
    'Metric': ['Train AUC (in-sample)', 'Validation AUC (out-of-sample)'],
    'Value': [f'{bench_train_auc:.4f}', f'{bench_val_auc:.4f}'],
})
print_table(bench_df, caption="Benchmark: Logistic Regression")
```

```
**Benchmark: Logistic Regression**
```

Metric	Value
Train AUC (in-sample)	0.9045
Validation AUC (out-of-sample)	0.8914

The benchmark sets a minimum threshold. Any candidate model that cannot beat the logistic regression validation AUC does not earn its complexity cost.

Candidate Model Tournament

Five algorithms tuned with **150-iteration randomized search** and 5-fold stratified CV. SMOTE oversampling is applied **inside each CV fold** via `imblearn.Pipeline` to

prevent synthetic-sample leakage across folds.

```
In [19]: #| label: model-training
        #| output: false

        # Define candidate models and hyperparameter search spaces
        models = {}
        param_grids = {}
        pos_weight = (
            (y_train == 0).sum() / (y_train == 1).sum()
        )

        if CATBOOST_AVAILABLE:
            models['CatBoost'] = SklearnCatBoost(
                random_state=RANDOM_STATE,
                verbose=0, thread_count=1,
            )
            param_grids['CatBoost'] = {
                'depth': [4, 6, 8, 10],
                'learning_rate': [0.01, 0.03, 0.05, 0.1],
                'iterations': [300, 500, 800, 1000],
                'l2_leaf_reg': [1, 3, 5, 7, 10],
                'border_count': [32, 64, 128, 254],
            }

        if XGBOOST_AVAILABLE:
            models['XGBoost'] = XGBClassifier(
                random_state=RANDOM_STATE,
                n_jobs=1, eval_metric='logloss',
            )
            param_grids['XGBoost'] = {
                'max_depth': [4, 6, 8, 10, 15],
                'learning_rate': [0.01, 0.03, 0.05, 0.1],
                'n_estimators': [300, 500, 800, 1000],
                'scale_pos_weight': [1, pos_weight],
                'subsample': [0.5, 0.7, 0.8, 0.9, 1.0],
                'colsample_bytree': [0.5, 0.7, 0.8, 0.9, 1.0],
                'reg_alpha': [0, 0.1, 0.5, 1.0],
                'reg_lambda': [1, 2, 5, 10],
                'min_child_weight': [1, 5, 10, 20],
            }

        if LIGHTGBM_AVAILABLE:
            models['LightGBM'] = LGBMClassifier(
                random_state=RANDOM_STATE,
                n_jobs=1, verbose=-1,
            )
            param_grids['LightGBM'] = {
                'num_leaves': [31, 63, 127, 255],
                'learning_rate': [0.01, 0.03, 0.05, 0.1],
                'n_estimators': [300, 500, 800, 1000],
                'class_weight': ['balanced', None],
                'subsample': [0.5, 0.7, 0.8, 0.9, 1.0],
                'colsample_bytree': [0.5, 0.7, 0.8, 0.9, 1.0],
                'reg_alpha': [0, 0.1, 0.5, 1.0],
```

```

        'reg_lambda': [1, 2, 5, 10],
        'min_child_samples': [5, 10, 20, 50],
    }

models['GradientBoosting'] = GradientBoostingClassifier(
    random_state=RANDOM_STATE,
)
param_grids['GradientBoosting'] = {
    'n_estimators': [100, 200, 300, 500],
    'max_depth': [4, 6, 8, 10, 15],
    'learning_rate': [0.01, 0.05, 0.1, 0.2],
    'subsample': [0.7, 0.8, 0.9, 1.0],
    'min_samples_leaf': [5, 10, 20],
}

models['RandomForest'] = RandomForestClassifier(
    random_state=RANDOM_STATE,
    n_jobs=1, class_weight='balanced',
)
param_grids['RandomForest'] = {
    'n_estimators': [200, 300, 500],
    'max_depth': [10, 15, 20, 25, None],
    'min_samples_split': [2, 5, 10],
    'min_samples_leaf': [1, 2, 4, 8],
}

cv = StratifiedKFold(
    n_splits=CV_FOLDS, shuffle=True,
    random_state=RANDOM_STATE,
)
best_models = {}
cv_results = {}

# Wrap each model in an imblearn Pipeline so SMOTE runs inside each CV fold
for name, model in models.items():
    if SMOTE_AVAILABLE:
        pipe = ImbPipeline([
            ('smote', SMOTE(random_state=RANDOM_STATE, k_neighbors=5)),
            ('model', model),
        ])
        prefixed_params = {
            f'model__{k}': v for k, v in param_grids[name].items()
        }
    else:
        pipe = model
        prefixed_params = param_grids[name]

    search = RandomizedSearchCV(
        estimator=pipe,
        param_distributions=prefixed_params,
        n_iter=N_ITER_SEARCH, cv=cv,
        scoring='roc_auc',
        n_jobs=N_JOBS,
        random_state=RANDOM_STATE, verbose=0,
    )
    search.fit(X_train_te, y_train)

```

```

# Extract the fitted model from the pipeline for downstream compatibility
if SMOTE_AVAILABLE:
    best_models[name] = search.best_estimator_.named_steps['model']
    cv_results[name] = {
        'best_score': search.best_score_,
        'best_params': {
            k.replace('model__', ''): v
            for k, v in search.best_params_.items()
        },
    }
else:
    best_models[name] = search.best_estimator_
    cv_results[name] = {
        'best_score': search.best_score_,
        'best_params': search.best_params_,
    }

# Evaluate each tuned model on the held-out validation set
val_results = {}
for name, model in best_models.items():
    probs = model.predict_proba(X_val_te)[: , 1]
    val_results[name] = {
        'auc': roc_auc_score(y_val, probs),
        'probs': probs,
    }

val_ranking = sorted(
    val_results.items(),
    key=lambda x: x[1]['auc'], reverse=True,
)

```

Model Comparison: CV vs. Validation AUC

```

In [20]: #| label: model-comparison-viz

fig, axes = plt.subplots(1, 2, figsize=(7.5, 3.8))

model_names = [name for name, _ in val_ranking]
cv_aucs = [
    cv_results[n]['best_score'] for n in model_names
]
val_aucs = [
    val_results[n]['auc'] for n in model_names
]

x = np.arange(len(model_names))
w = 0.35
axes[0].bar(
    x - w / 2, cv_aucs, w,
    label='CV AUC (5-fold)',
    color=COL_ACCENT, edgecolor='white',
)
axes[0].bar(
    x + w / 2, val_aucs, w,

```

```

        label='Validation AUC',
        color=COL_SUCCESS, edgecolor='white',
    )
    for i in range(len(model_names)):
        axes[0].text(
            x[i] - w / 2, cv_aucs[i] + 0.002,
            f'{cv_aucs[i]:.3f}',
            ha='center', fontsize=6, fontweight='bold',
        )
        axes[0].text(
            x[i] + w / 2, val_aucs[i] + 0.002,
            f'{val_aucs[i]:.3f}',
            ha='center', fontsize=6, fontweight='bold',
        )
    axes[0].set_xticks(x)
    axes[0].set_xticklabels(
        model_names, rotation=25, ha='right', fontsize=7,
    )
    axes[0].set_ylabel('AUC-ROC', fontsize=9)
    axes[0].set_title(
        'Model Tournament',
        fontweight='bold', fontsize=10,
    )
    axes[0].legend(fontsize=7)
    axes[0].set_ylim(
        min(cv_aucs + val_aucs) - 0.03,
        max(cv_aucs + val_aucs) + 0.02,
    )
    sns.despine(ax=axes[0])

    gaps = [
        cv_aucs[i] - val_aucs[i]
        for i in range(len(model_names))
    ]
    gap_colors = [
        COL_SUCCESS if abs(g) < 0.02
        else (COL_GOLD if abs(g) < 0.05 else COL_RISK)
        for g in gaps
    ]
    axes[1].barh(
        model_names, gaps,
        color=gap_colors, edgecolor='white',
    )
    for i, g in enumerate(gaps):
        axes[1].text(
            g + 0.001 * (1 if g >= 0 else -1), i,
            f'{g:+.4f}',
            va='center', fontsize=7, fontweight='bold',
        )
    axes[1].axvline(x=0, color='black', linewidth=0.8)
    axes[1].axvline(
        x=0.05, color=COL_RISK,
        linestyle='--', alpha=0.5, label='Overfit',
    )
    axes[1].axvline(
        x=-0.05, color=COL_RISK,

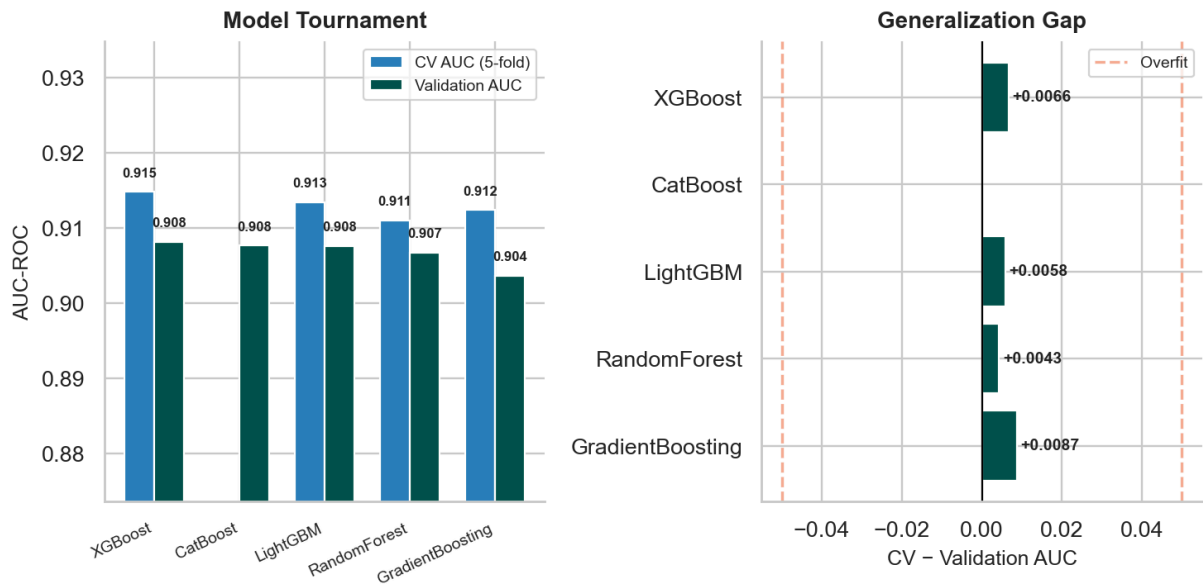
```

```

linestyle='--', alpha=0.5,
)
axes[1].set_xlabel('CV - Validation AUC', fontsize=9)
axes[1].set_title(
    'Generalization Gap',
    fontweight='bold', fontsize=10,
)
axes[1].legend(fontsize=7)
axes[1].invert_yaxis()
sns.despine(ax=axes[1])

plt.tight_layout()
plt.show()

```



Stacking & Voting Ensembles

Top three models serve as base learners in both a **stacking ensemble** (with LightGBM meta-learner) and a **soft voting ensemble**. The best performer is selected as champion.

```

In [21]: #| label: stacking
         #| output: false

         # Build ensembles from the top 3 individual models
         top_3_names = [name for name, _ in val_ranking[:3]]

         # Stacking with LightGBM meta-learner
         if LIGHTGBM_AVAILABLE:
             meta_learner = LGBMClassifier(
                 n_estimators=100, max_depth=3,
                 learning_rate=0.05, verbose=-1,
                 random_state=RANDOM_STATE, n_jobs=1,
             )
         else:
             meta_learner = LogisticRegression(

```

```

        random_state=RANDOM_STATE,
        max_iter=1000, class_weight='balanced',
    )

stacking_clf = StackingClassifier(
    estimators=[
        (name, best_models[name])
        for name in top_3_names
    ],
    final_estimator=meta_learner,
    cv=CV_FOLDS, stack_method='predict_proba',
    n_jobs=N_JOBS, passthrough=False,
)
stacking_clf.fit(X_train_te, y_train)
stack_val_probs = (
    stacking_clf.predict_proba(X_val_te)[:, 1]
)
stack_val_auc = roc_auc_score(y_val, stack_val_probs)

best_models['StackingEnsemble'] = stacking_clf
val_results['StackingEnsemble'] = {
    'auc': stack_val_auc, 'probs': stack_val_probs,
}

# --- Soft Voting ensemble ---
voting_clf = VotingClassifier(
    estimators=[
        (name, best_models[name])
        for name in top_3_names
    ],
    voting='soft',
)
voting_clf.fit(X_train_te, y_train)
vote_val_probs = (
    voting_clf.predict_proba(X_val_te)[:, 1]
)
vote_val_auc = roc_auc_score(y_val, vote_val_probs)

best_models['VotingEnsemble'] = voting_clf
val_results['VotingEnsemble'] = {
    'auc': vote_val_auc, 'probs': vote_val_probs,
}

# --- Select champion ---
champion_name = max(
    val_results.items(),
    key=lambda x: x[1]['auc'],
)[0]
champion_model = best_models[champion_name]

# Re-sort ranking including ensembles
val_ranking = sorted(
    val_results.items(),
    key=lambda x: x[1]['auc'], reverse=True,
)

```

Ensemble Waterfall: Additive AUC

```
In [22]: #| label: ensemble-waterfall

baseline_auc = 0.500
individual_aucs = [
    (name, val_results[name]['auc'])
    for name in top_3_names
]
individual_aucs_sorted = sorted(
    individual_aucs, key=lambda x: x[1],
)
ensemble_auc = stack_val_auc

wf_labels = ['Random\nBaseline']
wf_bottoms = [0]
wf_heights = [baseline_auc]
wf_colors = [COL_NEUTRAL]

running = baseline_auc
for i, (name, auc) in enumerate(
    individual_aucs_sorted
):
    if i == 0:
        inc = auc - baseline_auc
    else:
        inc = auc - individual_aucs_sorted[i - 1][1]
    wf_labels.append(f'{name}\n({auc:.3f})')
    wf_bottoms.append(running)
    wf_heights.append(inc)
    wf_colors.append(
        MODEL_COLORS.get(name, COL_ACCENT)
    )
    running += inc

ens_inc = ensemble_auc - individual_aucs_sorted[-1][1]
wf_labels.append('Ensemble\nBonus')
wf_bottoms.append(running)
wf_heights.append(ens_inc)
wf_colors.append(COL_GOLD)
running += ens_inc

wf_labels.append(
    f'Stacking\n({ensemble_auc:.3f})'
)
wf_bottoms.append(0)
wf_heights.append(ensemble_auc)
wf_colors.append(COL_SUCCESS)

fig, ax = plt.subplots(figsize=(7.5, 4))
x_pos = np.arange(len(wf_labels))
ax.bar(
    x_pos, wf_heights, bottom=wf_bottoms,
    color=wf_colors, edgecolor='white',
```

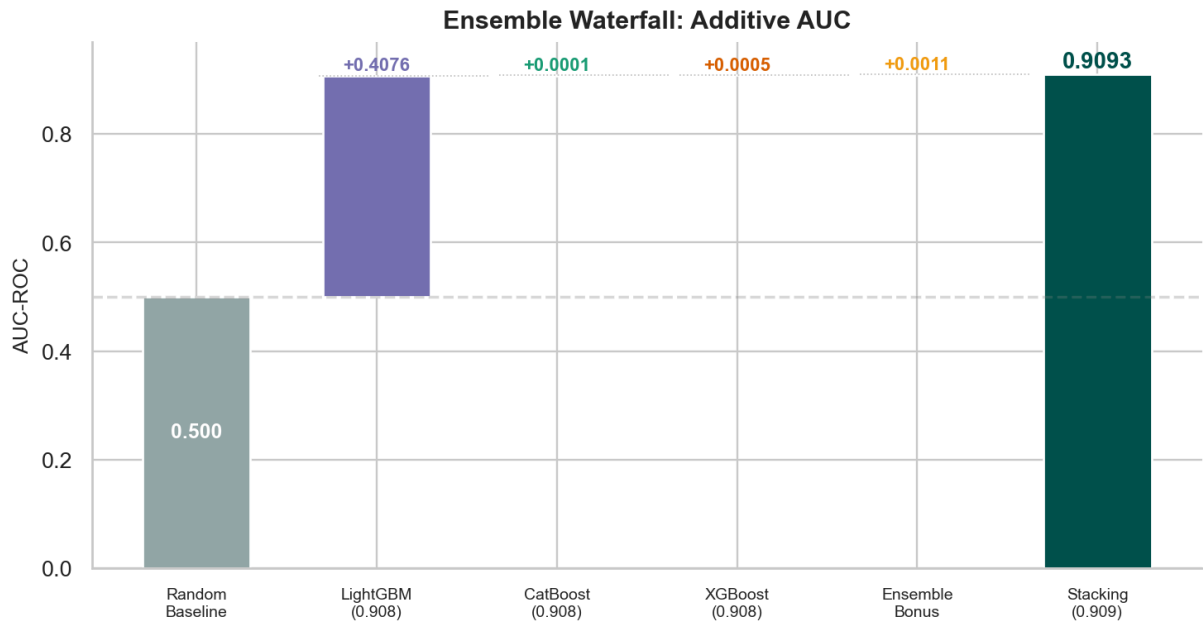
```

        linewidth=1, width=0.6,
    )
    for i, (h, bot) in enumerate(
        zip(wf_heights, wf_bottoms)
    ):
        if i == 0:
            ax.text(
                i, bot + h / 2, f'{h:.3f}',
                ha='center', va='center',
                fontweight='bold', fontsize=9, color='white',
            )
        elif i == len(wf_labels) - 1:
            ax.text(
                i, h + 0.003, f'{h:.4f}',
                ha='center', va='bottom',
                fontweight='bold', fontsize=10,
                color=COL_SUCCESS,
            )
        else:
            ax.text(
                i, bot + h + 0.003, f'+{h:.4f}',
                ha='center', va='bottom',
                fontweight='bold', fontsize=8,
                color=wf_colors[i],
            )

    for i in range(1, len(wf_labels) - 1):
        top = wf_bottoms[i] + wf_heights[i]
        ax.plot(
            [i - 0.32, i + 0.62], [top, top],
            color='gray', linewidth=0.6,
            linestyle=':', alpha=0.5,
        )

    ax.set_xticks(x_pos)
    ax.set_xticklabels(wf_labels, fontsize=7)
    ax.set_ylabel('AUC-ROC', fontsize=9)
    ax.set_title(
        'Ensemble Waterfall: Additive AUC',
        fontweight='bold', fontsize=11,
    )
    ax.set_ylim(0, ensemble_auc + 0.06)
    ax.axhline(
        y=baseline_auc, color='gray',
        linestyle='--', alpha=0.3,
    )
    sns.despine()
    plt.tight_layout()
    plt.show()

```

Ablation Studies

```
In [23]: #| label: ablation-studies

feature_groups = {
    'Channel': [
        c for c in X_train_te.columns
        if 'channel' in c.lower()
    ],
    'Intent': [
        c for c in X_train_te.columns
        if 'intent' in c.lower()
    ],
    'Title/Seniority': [
        c for c in X_train_te.columns
        if any(
            t in c.lower()
            for t in [
                'seniority', 'function', 'scope',
                'decision', 'title_word',
            ]
        )
    ],
    'Capital Density': [
        c for c in X_train_te.columns
        if 'capital' in c.lower()
    ],
    'Hidden Gem': [
        c for c in X_train_te.columns
        if 'hidden_gem' in c.lower()
    ],
    'Role-Product': [
        c for c in X_train_te.columns
        if 'role_product' in c.lower()
    ],
}
```

```

'Title Bigrams': [
    c for c in X_train_te.columns
    if c.startswith('has_') or 'bigram' in c.lower()
],
'LSA Text': [
    c for c in X_train_te.columns
    if c.startswith('lsa_')
],
'Interactions': [
    c for c in X_train_te.columns
    if any(
        t in c.lower()
        for t in [
            '_x_', 'power_trio', 'golden',
            'senior_pharma', 'global_scope',
            'industry_channel', 'decision_relevant',
        ]
    )
],
'Temporal': [
    c for c in X_train_te.columns
    if any(
        t in c.lower()
        for t in [
            'lead_age', 'velocity',
            'fresh', 'stale', 'decay',
        ]
    )
],
]
}
feature_groups = {
    k: v for k, v in feature_groups.items()
    if len(v) > 0
}

ablation_models = {
    name: best_models[name]
    for name in top_3_names
    if name in best_models
}

ablation_results = {}
for model_name, model in ablation_models.items():
    full_auc = val_results[model_name]['auc']
    drops = {}
    for grp_name, grp_cols in feature_groups.items():
        keep = [
            c for c in X_train_te.columns
            if c not in grp_cols
        ]
        if len(keep) == len(X_train_te.columns):
            continue
        try:
            abl = clone(model)
            abl.fit(X_train_te[keep], y_train)
            abl_probs = (

```

```

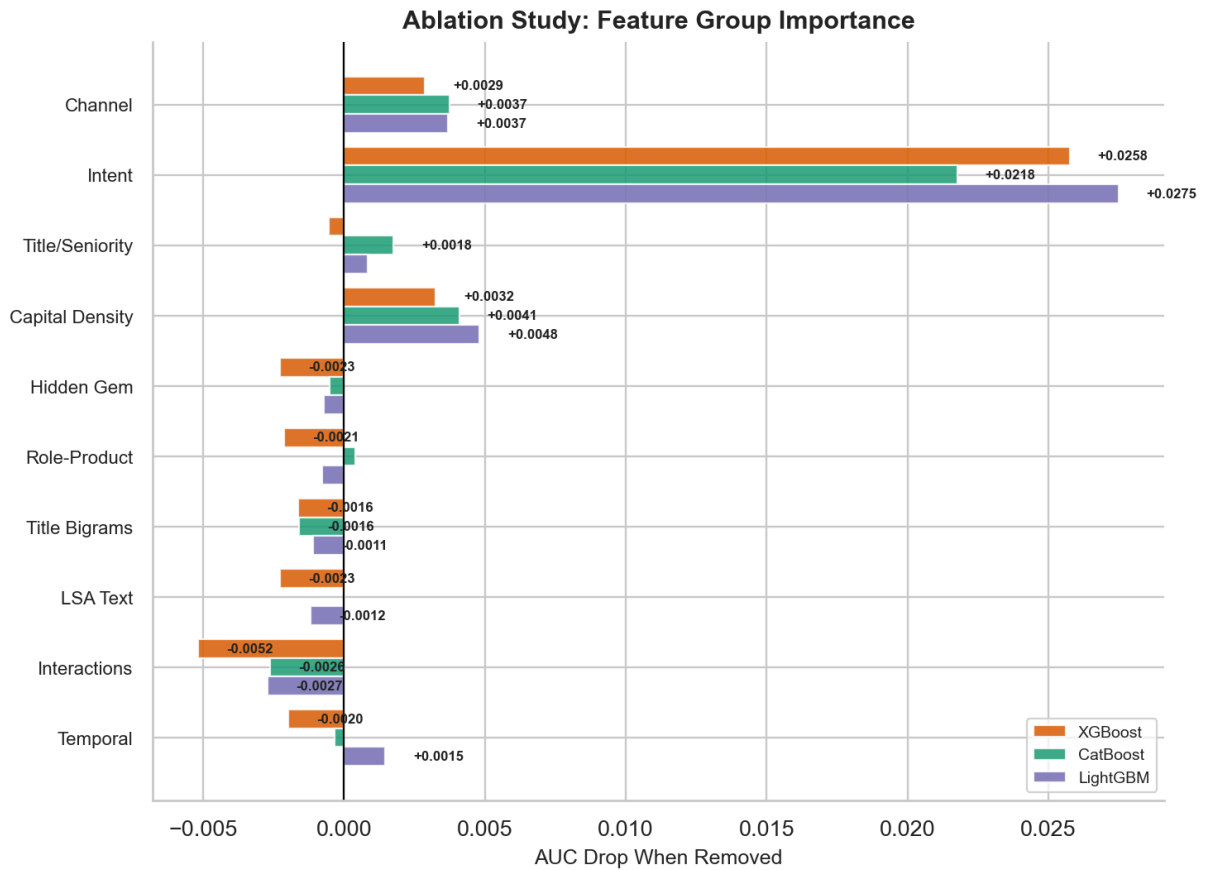
        abl.predict_proba(X_val_te[keep])[:, 1]
    )
    abl_auc = roc_auc_score(y_val, abl_probs)
    drops[grp_name] = full_auc - abl_auc
    except Exception:
        drops[grp_name] = 0.0
    ablation_results[model_name] = drops

fig, ax = plt.subplots(
    figsize=(7.5, max(4, len(feature_groups) * 0.55))
)
group_names = list(feature_groups.keys())
n_m = len(ablation_results)
bh = 0.8 / n_m
y_pos = np.arange(len(group_names))

for i, (mn, drops) in enumerate(
    ablation_results.items()
):
    vals = [drops.get(g, 0) for g in group_names]
    color = MODEL_COLORS.get(mn, COL_ACCENT)
    offset = (i - n_m / 2 + 0.5) * bh
    ax.barh(
        y_pos + offset, vals, height=bh,
        label=mn, color=color,
        edgecolor='white', alpha=0.85,
    )
    for j, v in enumerate(vals):
        if abs(v) > 0.001:
            ax.text(
                v + 0.001, y_pos[j] + offset,
                f'{v:+.4f}',
                va='center', fontsize=6,
                fontweight='bold',
            )

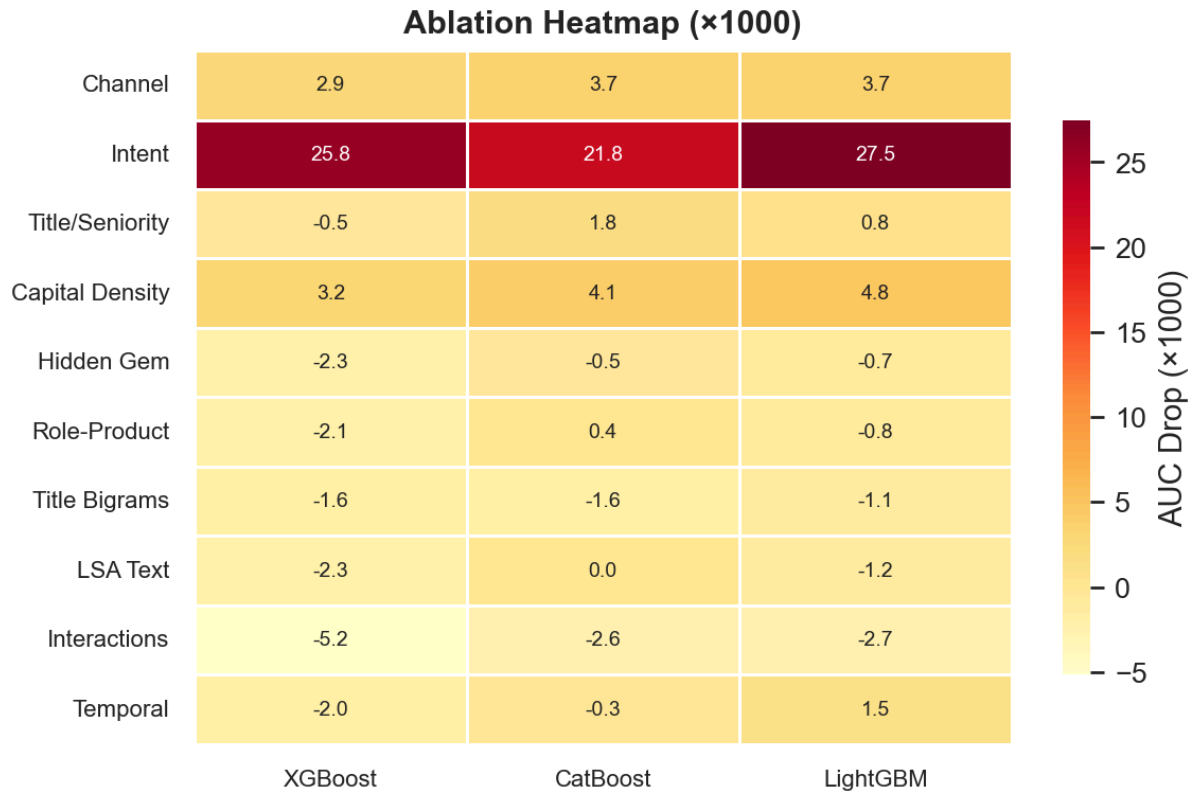
ax.set_yticks(y_pos)
ax.set_yticklabels(group_names, fontsize=8)
ax.set_xlabel(
    'AUC Drop When Removed', fontsize=9,
)
ax.set_title(
    'Ablation Study: Feature Group Importance',
    fontweight='bold', fontsize=11,
)
ax.axvline(x=0, color='black', linewidth=0.8)
ax.legend(loc='lower right', fontsize=7)
ax.invert_yaxis()
sns.despine()
plt.tight_layout()
plt.show()

```



```
In [24]: #| label: ablation-heatmap

ablation_df = (
    pd.DataFrame(ablation_results)
    .reindex(group_names).fillna(0)
)
fig, ax = plt.subplots(figsize=(6, 4))
sns.heatmap(
    ablation_df * 1000, annot=True, fmt='.1f',
    cmap='YlOrRd', ax=ax, linewidths=0.5,
    cbar_kws={
        'label': 'AUC Drop (×1000)', 'shrink': 0.8,
    },
    annot_kws={'size': 7},
)
ax.set_title(
    'Ablation Heatmap (×1000)',
    fontweight='bold', fontsize=11,
)
ax.set_ylabel('')
ax.set_xlabel('')
ax.tick_params(labelsize=8)
plt.tight_layout()
plt.show()
```



Test Set Evaluation

```
In [25]: #| label: test-evaluation
        #| output: false

        # Final evaluation on held-out test set (never seen during training or selection)
        test_probs = champion_model.predict_proba(
            X_test_te
        )[:, 1]
        test_preds = (test_probs >= 0.5).astype(int)

        test_auc = roc_auc_score(y_test, test_probs)
        test_ap = average_precision_score(y_test, test_probs)
        test_brier = brier_score_loss(y_test, test_probs)
        test_logloss = log_loss(y_test, test_probs)

        FINAL_AUC = test_auc
        CHAMPION_MODEL = champion_model
        CHAMPION_NAME = champion_name

        # In-sample metrics for generalization comparison
        train_probs = champion_model.predict_proba(X_train_te)[:, 1]
        train_auc = roc_auc_score(y_train, train_probs)
        train_ap = average_precision_score(y_train, train_probs)
        train_brier = brier_score_loss(y_train, train_probs)
        train_logloss = log_loss(y_train, train_probs)
```

In-Sample vs. Out-of-Sample Performance

Comparing training and test metrics reveals whether the champion model generalizes or has memorized the training data.

```
In [26]: #| label: insample-vs-oosample

# Side-by-side generalization check
gen_df = pd.DataFrame({
    'Metric': ['AUC-ROC', 'Average Precision', 'Brier Score', 'Log Loss'],
    'Train (In-Sample)': [
        f'{train_auc:.4f}', f'{train_ap:.4f}',
        f'{train_brier:.4f}', f'{train_logloss:.4f}',
    ],
    'Test (Out-of-Sample)': [
        f'{test_auc:.4f}', f'{test_ap:.4f}',
        f'{test_brier:.4f}', f'{test_logloss:.4f}',
    ],
    'Gap': [
        f'{train_auc - test_auc:+.4f}',
        f'{train_ap - test_ap:+.4f}',
        f'{test_brier - train_brier:+.4f}',
        f'{test_logloss - train_logloss:+.4f}',
    ],
})
print_table(
    gen_df,
    caption=f"Generalization Check -{CHAMPION_NAME}",
)
```

****Generalization Check -VotingEnsemble****

Metric	Train (In-Sample)	Test (Out-of-Sample)	Gap
AUC-ROC	0.9927	0.9149	0.0778
Average Precision	0.9767	0.7035	0.2732
Brier Score	0.0344	0.0853	0.0509
Log Loss	0.1408	0.2748	0.134

A small positive gap in AUC (train > test) is normal. Gaps above 0.05 would signal overfitting. The Brier and Log Loss gaps confirm probability calibration holds on unseen data.

```
In [27]: #| label: test-metrics

metrics_df = pd.DataFrame({
    'Metric': [
        'AUC-ROC', 'Average Precision',
        'Brier Score', 'Log Loss',
    ],
})
```

```

    'Value': [
        f'{test_auc:.4f}', f'{test_ap:.4f}',
        f'{test_brier:.4f}', f'{test_logloss:.4f}',
    ],
    'Interpretation': [
        'Discrimination -higher is better',
        'Precision-recall -higher is better',
        'Calibration error -lower is better',
        'Probabilistic loss -lower is better',
    ],
})
print_table(
    metrics_df,
    caption=f"Test Set Performance -{CHAMPION_NAME}",
)

```

****Test Set Performance -VotingEnsemble****

Metric	Value	Interpretation
AUC-ROC	0.9149	Discrimination -higher is better
Average Precision	0.7035	Precision-recall -higher is better
Brier Score	0.0853	Calibration error -lower is better
Log Loss	0.2748	Probabilistic loss -lower is better

ROC & Precision-Recall Curves

```

In [28]: #| label: roc-all-models

fig, axes = plt.subplots(1, 2, figsize=(7.5, 4))

for name in list(best_models.keys()):
    probs = (
        best_models[name]
        .predict_proba(X_test_te)[: , 1]
    )
    fpr, tpr, _ = roc_curve(y_test, probs)
    auc_v = roc_auc_score(y_test, probs)
    lw = 2.5 if name == CHAMPION_NAME else 1
    alpha = 1.0 if name == CHAMPION_NAME else 0.5
    axes[0].plot(
        fpr, tpr,
        color=MODEL_COLORS.get(name, COL_NEUTRAL),
        linewidth=lw, alpha=alpha,
        label=f'{name} ({auc_v:.3f})',
    )

    axes[0].plot(
        [0, 1], [0, 1], 'k--', alpha=0.3,
        label='Random',
    )

axes[0].set_xlabel('FPR', fontsize=9)
axes[0].set_ylabel('TPR', fontsize=9)
axes[0].set_title(
    'ROC Curves', fontweight='bold', fontsize=10,

```

```

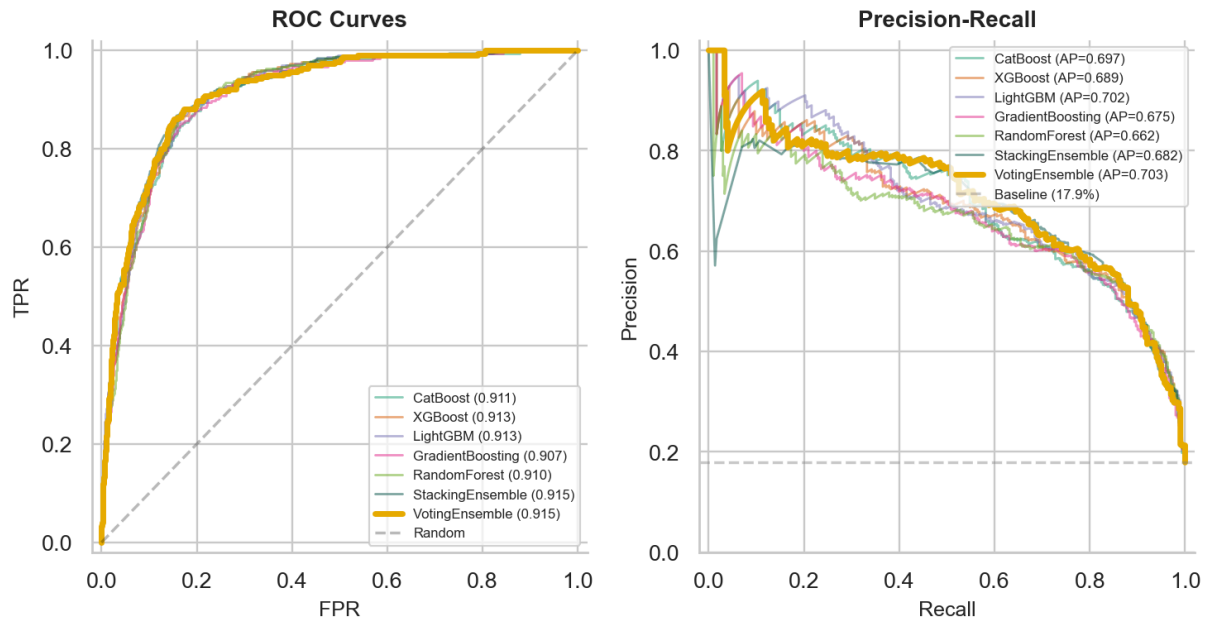
)
axes[0].legend(loc='lower right', fontsize=6)
axes[0].set_xlim(-0.02, 1.02)
axes[0].set_ylim(-0.02, 1.02)
sns.despine(ax=axes[0])

for name in list(best_models.keys()):
    probs = (
        best_models[name]
        .predict_proba(X_test_te)[: , 1]
    )
    prec, rec, _ = precision_recall_curve(
        y_test, probs,
    )
    ap = average_precision_score(y_test, probs)
    lw = 2.5 if name == CHAMPION_NAME else 1
    alpha = 1.0 if name == CHAMPION_NAME else 0.5
    axes[1].plot(
        rec, prec,
        color=MODEL_COLORS.get(name, COL_NEUTRAL),
        linewidth=lw, alpha=alpha,
        label=f'{name} (AP={ap:.3f})',
    )

axes[1].axhline(
    y=y_test.mean(), color='gray',
    linestyle='--', alpha=0.4,
    label=f'Baseline ({y_test.mean():.1%})',
)
axes[1].set_xlabel('Recall', fontsize=9)
axes[1].set_ylabel('Precision', fontsize=9)
axes[1].set_title(
    'Precision-Recall', fontweight='bold', fontsize=10,
)
axes[1].legend(loc='upper right', fontsize=6)
axes[1].set_xlim(-0.02, 1.02)
axes[1].set_ylim(0, 1.02)
sns.despine(ax=axes[1])

plt.tight_layout()
plt.show()

```

Calibration & Score Distribution

```
In [29]: #| label: calibration-and-scores

fig, axes = plt.subplots(1, 2, figsize=(7.5, 3.8))

prob_true, prob_pred = calibration_curve(
    y_test, test_probs, n_bins=10,
    strategy='uniform',
)
axes[0].plot(
    prob_pred, prob_true, 's-',
    color=COL_SUCCESS, linewidth=2,
    markersize=6, label=CHAMPION_NAME,
)
axes[0].plot(
    [0, 1], [0, 1], 'k--', alpha=0.4,
    label='Perfect',
)
axes[0].fill_between(
    prob_pred, prob_pred, prob_true,
    alpha=0.15, color=COL_SUCCESS,
)
axes[0].set_xlabel('Predicted Prob.', fontsize=9)
axes[0].set_ylabel('Actual Fraction', fontsize=9)
axes[0].set_title(
    'Calibration Curve',
    fontweight='bold', fontsize=10,
)
axes[0].legend(fontsize=7)
axes[0].set_xlim(-0.02, 1.02)
axes[0].set_ylim(-0.02, 1.02)
sns.despine(ax=axes[0])

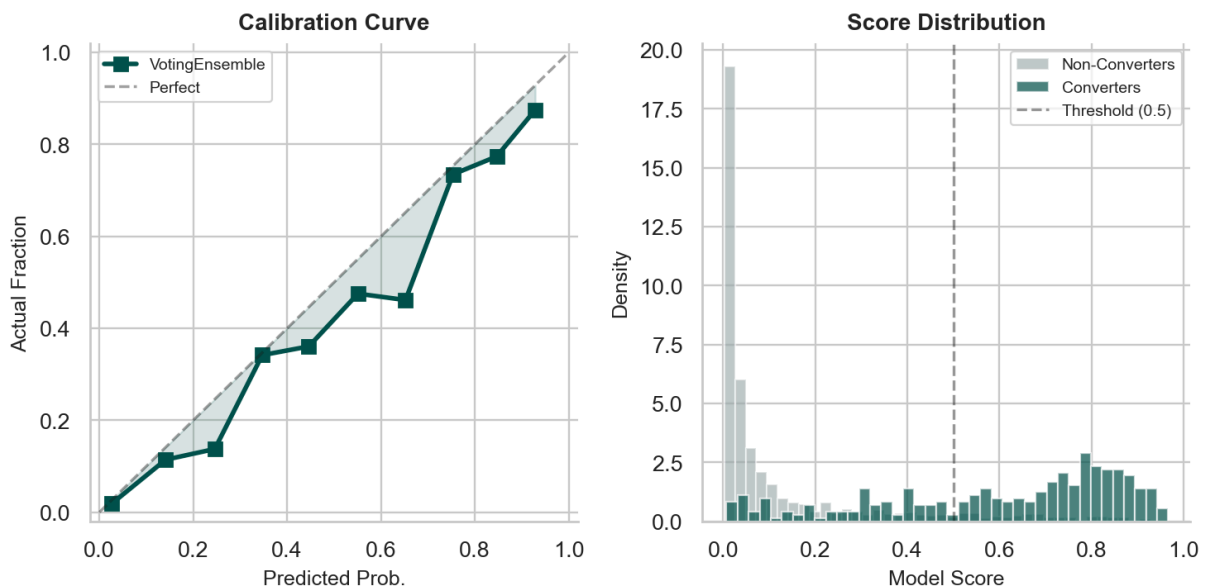
axes[1].hist(
```

```

test_probs[y_test == 0], bins=40, alpha=0.6,
color=COL_NEUTRAL, label='Non-Converters',
density=True,
)
axes[1].hist(
    test_probs[y_test == 1], bins=40, alpha=0.7,
    color=COL_SUCCESS, label='Converters',
    density=True,
)
axes[1].axvline(
    x=0.5, color='black', linestyle='--', alpha=0.4,
    label='Threshold (0.5)',
)
axes[1].set_xlabel('Model Score', fontsize=9)
axes[1].set_ylabel('Density', fontsize=9)
axes[1].set_title(
    'Score Distribution',
    fontweight='bold', fontsize=10,
)
axes[1].legend(fontsize=7)
sns.despine(ax=axes[1])

plt.tight_layout()
plt.show()

```



Confusion Matrix

```

In [30]: #| label: confusion-matrix

fig, axes = plt.subplots(1, 2, figsize=(7, 3.5))

cm = confusion_matrix(y_test, test_preds)
sns.heatmap(
    cm, annot=True, fmt='d', cmap='Blues',
    ax=axes[0],
    xticklabels=['Pred: No', 'Pred: Yes'],

```

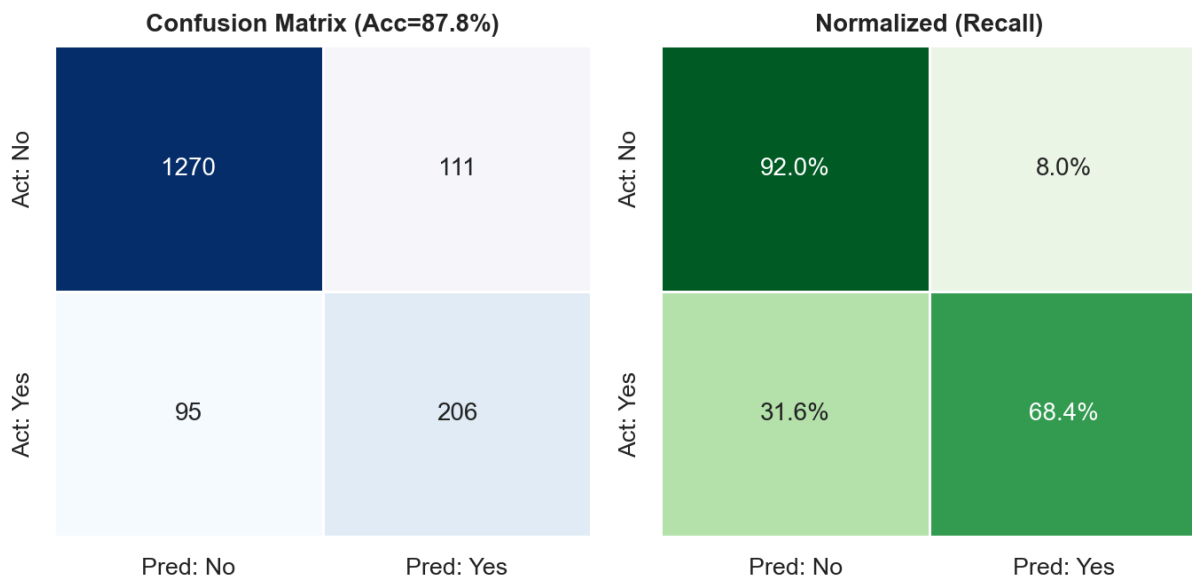
```

yticklabels=['Act: No', 'Act: Yes'],
linewidths=1, cbar=False,
annot_kws={'size': 10},
)
acc = (cm[0, 0] + cm[1, 1]) / cm.sum()
axes[0].set_title(
    f'Confusion Matrix (Acc={acc:.1%})',
    fontweight='bold', fontsize=10,
)

cm_norm = (
    cm.astype(float)
    / cm.sum(axis=1, keepdims=True)
)
sns.heatmap(
    cm_norm, annot=True, fmt='.1%', cmap='Greens',
    ax=axes[1],
    xticklabels=['Pred: No', 'Pred: Yes'],
    yticklabels=['Act: No', 'Act: Yes'],
    linewidths=1, cbar=False, vmin=0, vmax=1,
    annot_kws={'size': 10},
)
axes[1].set_title(
    'Normalized (Recall)',
    fontweight='bold', fontsize=10,
)

plt.tight_layout()
plt.show()

```



SHAP Explainability

```

In [31]: #| label: shap-analysis
         #| output: false

         # SHAP explainability - uses TreeExplainer for tree models, KernelExplainer

```

```

shap_values = None
shap_model_name = CHAMPION_NAME
X_explain = None

if SHAP_AVAILABLE:
    np.random.seed(RANDOM_STATE)
    bg_idx = np.random.choice(
        len(X_train_te),
        min(SHAP_BACKGROUND_SAMPLES, len(X_train_te)),
        replace=False,
    )
    test_idx = np.random.choice(
        len(X_test_te),
        min(SHAP_TEST_SAMPLES, len(X_test_te)),
        replace=False,
    )
    X_bg = X_train_te.iloc[bg_idx]
    X_explain = X_test_te.iloc[test_idx]

    try:
        tree_models = [
            'CatBoost', 'XGBoost', 'LightGBM',
            'GradientBoosting', 'RandomForest',
        ]
        if CHAMPION_NAME in tree_models:
            base = (
                CHAMPION_MODEL._model
                if CHAMPION_NAME == 'CatBoost'
                and hasattr(CHAMPION_MODEL, '_model')
                else CHAMPION_MODEL
            )
            explainer = shap.TreeExplainer(base)
            shap_values = explainer.shap_values(
                X_explain,
            )
            if isinstance(shap_values, list):
                shap_values = shap_values[1]
        elif CHAMPION_NAME in [
            'StackingEnsemble', 'VotingEnsemble',
        ]:
            best_base_name = top_3_names[0]
            shap_model_name = (
                f'{best_base_name} (top base learner)'
            )
            best_base = best_models[best_base_name]
            base = (
                best_base._model
                if best_base_name == 'CatBoost'
                and hasattr(best_base, '_model')
                else best_base
            )
            explainer = shap.TreeExplainer(base)
            shap_values = explainer.shap_values(
                X_explain,
            )
            if isinstance(shap_values, list):

```

```

        shap_values = shap_values[1]
    else:
        explainer = shap.KernelExplainer(
            lambda x: (
                CHAMPION_MODEL.predict_proba(x)[:, 1]
            ),
            X_bg,
        )
        shap_values = explainer.shap_values(
            X_explain, nsamples=100,
        )
    except Exception as e:
        print(f"SHAP error: {e}")

```

In [32]: *#| label: shap-plots*

```

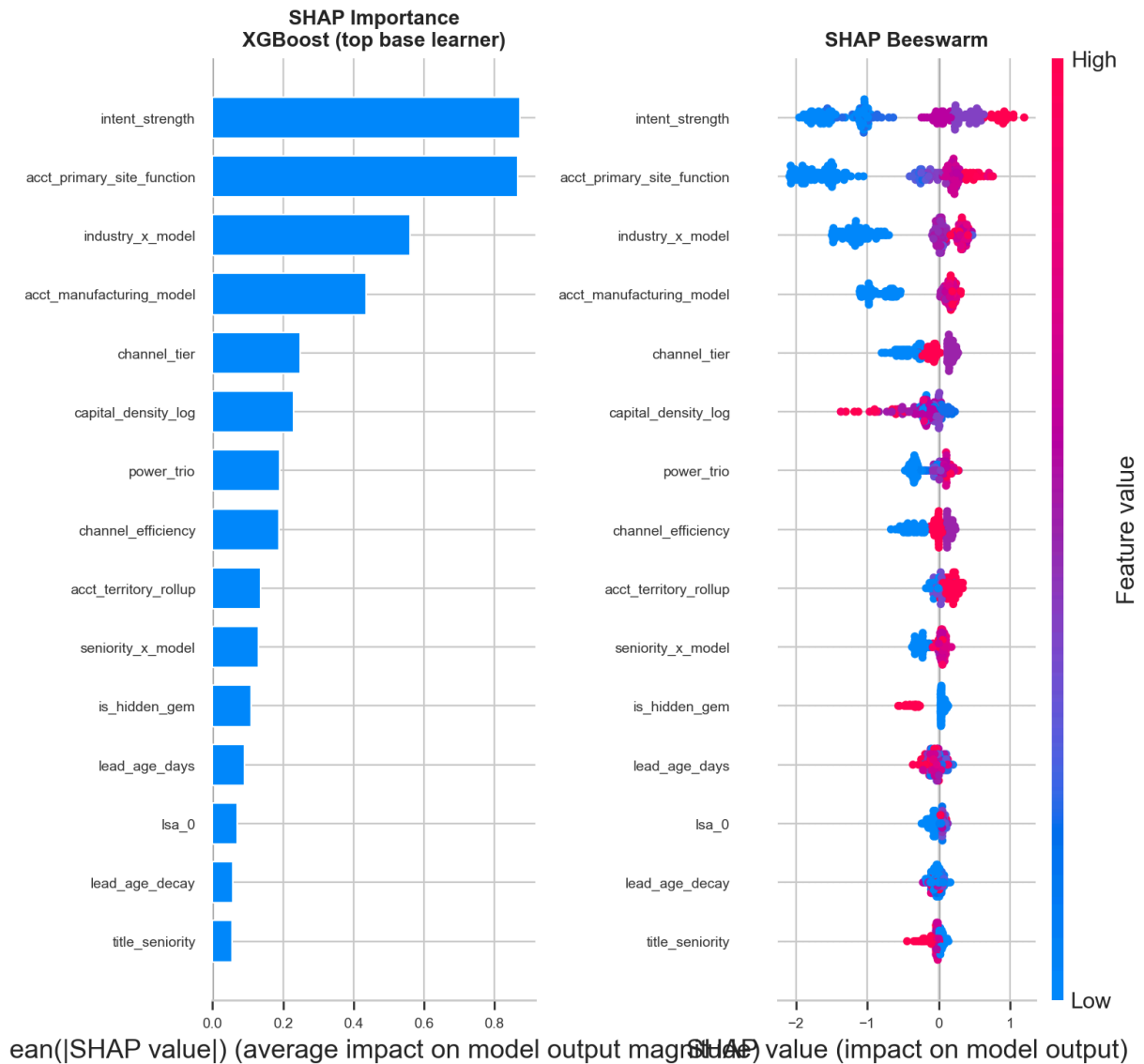
if SHAP_AVAILABLE and shap_values is not None:
    fig, axes = plt.subplots(1, 2, figsize=(7.5, 5))

    plt.sca(axes[0])
    shap.summary_plot(
        shap_values, X_explain,
        plot_type="bar", show=False, max_display=15,
    )
    axes[0].set_title(
        f'SHAP Importance\n{shap_model_name}',
        fontweight='bold', fontsize=10,
    )
    axes[0].tick_params(labelsize=7)

    plt.sca(axes[1])
    shap.summary_plot(
        shap_values, X_explain,
        show=False, max_display=15,
    )
    axes[1].set_title(
        'SHAP Beeswarm',
        fontweight='bold', fontsize=10,
    )
    axes[1].tick_params(labelsize=7)

    plt.tight_layout()
    plt.show()

```



ean(|SHAP value|) (average impact on model output magnitude) SHAP value (impact on model output)

SHAP Dependence Plots

```
In [33]: #| label: shap-dependence

if SHAP_AVAILABLE and shap_values is not None:
    mean_abs = np.abs(shap_values).mean(axis=0)
    top_idx = np.argsort(mean_abs)[-4:][::-1]
    top_feat = [
        X_explain.columns[i] for i in top_idx
    ]

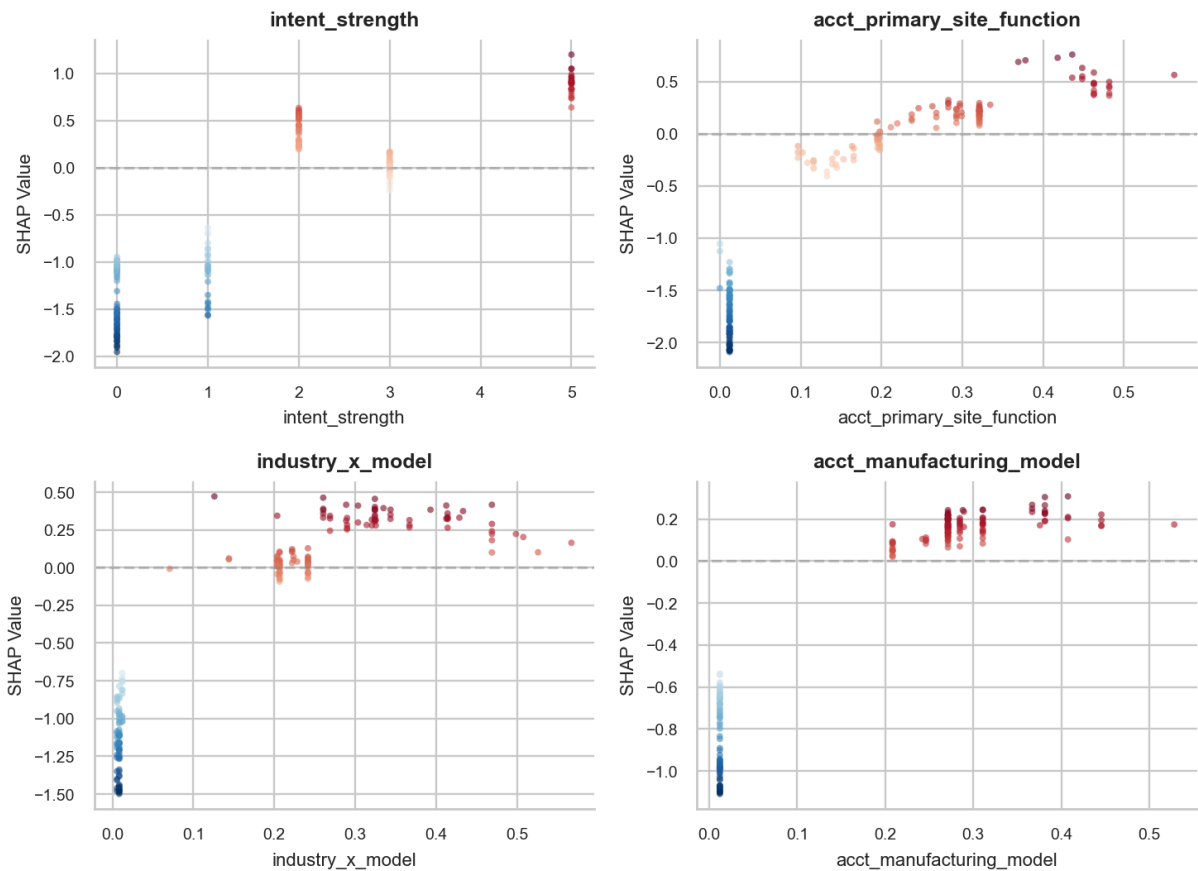
    fig, axes = plt.subplots(2, 2, figsize=(7.5, 5.5))
    for ax, fn, fi in zip(
        axes.flat, top_feat, top_idx
    ):
        ax.scatter(
            X_explain[fn].values,
            shap_values[:, fi],
            c=shap_values[:, fi], cmap='RdBu_r',
            s=8, alpha=0.6, edgecolors='none',
```

```

    )
    ax.axhline(
        y=0, color='gray',
        linestyle='--', alpha=0.4,
    )
    ax.set_xlabel(fn, fontsize=8)
    ax.set_ylabel('SHAP Value', fontsize=8)
    ax.set_title(
        f'{fn}', fontweight='bold', fontsize=9,
    )
    ax.tick_params(labelsize=7)
    sns.despine(ax=ax)

plt.tight_layout()
plt.show()

```



Profit Analysis

```

In [34]: #| label: profit-curve-calc
         #| output: false

         # Rank leads by predicted probability, compute cumulative profit at each threshold
         def calculate_profit_curve(
             y_true, y_probs,
             cost=COST_PER_CALL, value=VALUE_PER_SQL,
         ):
             order = np.argsort(y_probs)[::-1]

```

```

y_s = y_true[order]
p_s = y_probs[order]
n = len(y_true)
cs = np.cumsum(y_s)
results = []
for k in range(1, n + 1):
    results.append({
        'threshold': p_s[k - 1],
        'n_calls': k, 'n_sqls': cs[k - 1],
        'revenue': cs[k - 1] * value,
        'cost': k * cost,
        'profit': cs[k - 1] * value - k * cost,
        'pct_population': k / n,
        'pct_sqls_captured': (
            cs[k - 1] / y_true.sum()
            if y_true.sum() > 0 else 0
        ),
        'lift': (
            (cs[k - 1] / k)
            / (y_true.sum() / n)
            if k > 0 else 0
        ),
    })
return pd.DataFrame(results)

profit_df = calculate_profit_curve(y_test, test_probs)
opt = profit_df.iloc[profit_df['profit'].idxmax()]
OPTIMAL_THRESHOLD = opt['threshold']
MAX_PROFIT = opt['profit']
OPTIMAL_CALLS = opt['n_calls']
OPTIMAL_SQLS = opt['n_sqls']
OPTIMAL_PCT_POP = opt['pct_population']
OPTIMAL_PCT_CAPTURE = opt['pct_sqls_captured']

```

```

In [35]: #| label: profit-summary-table

profit_summary = pd.DataFrame({
    'Metric': [
        'Optimal Threshold', 'Leads to Contact',
        'SQLs Captured', '% Population',
        'SQL Capture Rate', 'Net Profit',
        'Lift vs Random',
    ],
    'Value': [
        f'{OPTIMAL_THRESHOLD:.3f}',
        f'{int(OPTIMAL_CALLS):,}',
        f'{int(OPTIMAL_SQLS):,}',
        f'{OPTIMAL_PCT_POP:.1%}',
        f'{OPTIMAL_PCT_CAPTURE:.1%}',
        f'${MAX_PROFIT:,.0f}',
        f'{OPTIMAL_PCT_CAPTURE / OPTIMAL_PCT_POP:.1f}x',
    ],
})
print_table(
    profit_summary,

```



```
caption="Optimal Profit Configuration",
)
```

****Optimal Profit Configuration****

Metric	Value
Optimal Threshold	0.027
Leads to Contact	1,041
SQLs Captured	298
% Population	61.9%
SQL Capture Rate	99.0%
Net Profit	\$1,735,950
Lift vs Random	1.6x

Profit, Gains, Lift & Decile

```
In [36]: #| label: profit-4panel

fig, axes = plt.subplots(2, 2, figsize=(7.5, 6))

ax1 = axes[0, 0]
ax1.plot(
    profit_df['pct_population'] * 100,
    profit_df['profit'] / 1000,
    color=COL_PROFIT, linewidth=2,
)
ax1.axvline(
    x=OPTIMAL_PCT_POP * 100, color=COL_GOLD,
    linestyle='--',
    label=f'Optimal: {OPTIMAL_PCT_POP:.1%}',
)
ax1.scatter(
    [OPTIMAL_PCT_POP * 100], [MAX_PROFIT / 1000],
    color=COL_GOLD, s=80, zorder=5, marker='*',
)
ax1.fill_between(
    profit_df['pct_population'] * 100, 0,
    profit_df['profit'] / 1000,
    alpha=0.12, color=COL_PROFIT,
)
ax1.set_xlabel('% Contacted', fontsize=8)
ax1.set_ylabel('Profit ($K)', fontsize=8)
ax1.set_title('Profit Curve', fontweight='bold', fontsize=9)
ax1.legend(fontsize=6)
ax1.set_xlim(0, 100)

ax2 = axes[0, 1]
ax2.plot(
    profit_df['pct_population'] * 100,
    profit_df['pct_sqls_captured'] * 100,
    color=COL_SUCCESS, linewidth=2, label='Model',
)
ax2.plot(
    [0, 100], [0, 100], 'k--', alpha=0.4,
```

```

        label='Random',
    )
    ax2.fill_between(
        profit_df['pct_population'] * 100,
        profit_df['pct_population'] * 100,
        profit_df['pct_sqls_captured'] * 100,
        alpha=0.12, color=COL_SUCCESS,
    )
    ax2.set_xlabel('% Contacted', fontsize=8)
    ax2.set_ylabel('% SQLs Captured', fontsize=8)
    ax2.set_title(
        'Cumulative Gains', fontweight='bold', fontsize=9,
    )
    ax2.legend(loc='lower right', fontsize=6)
    ax2.set_xlim(0, 100)
    ax2.set_ylim(0, 100)

    ax3 = axes[1, 0]
    ax3.plot(
        profit_df['pct_population'] * 100,
        profit_df['lift'],
        color=COL_ACCENT, linewidth=2,
    )
    ax3.axhline(
        y=1, color='gray', linestyle='--', alpha=0.4,
        label='Baseline (1.0x)',
    )
    ax3.fill_between(
        profit_df['pct_population'] * 100,
        1, profit_df['lift'],
        where=profit_df['lift'] > 1,
        alpha=0.12, color=COL_SUCCESS,
    )
    ax3.set_xlabel('% Contacted', fontsize=8)
    ax3.set_ylabel('Lift', fontsize=8)
    ax3.set_title('Lift Chart', fontweight='bold', fontsize=9)
    ax3.legend(fontsize=6)
    ax3.set_xlim(0, 100)

    ax4 = axes[1, 1]
    decile_profits = []
    for i in range(10):
        s, e = i * 0.1, (i + 1) * 0.1
        mask = (
            (profit_df['pct_population'] > s)
            & (profit_df['pct_population'] <= e)
        )
        if mask.any():
            row = profit_df[mask].iloc[-1]
            prev = (
                profit_df[
                    profit_df['pct_population'] <= s
                ].iloc[-1]['profit']
                if i > 0 else 0
            )
            decile_profits.append(

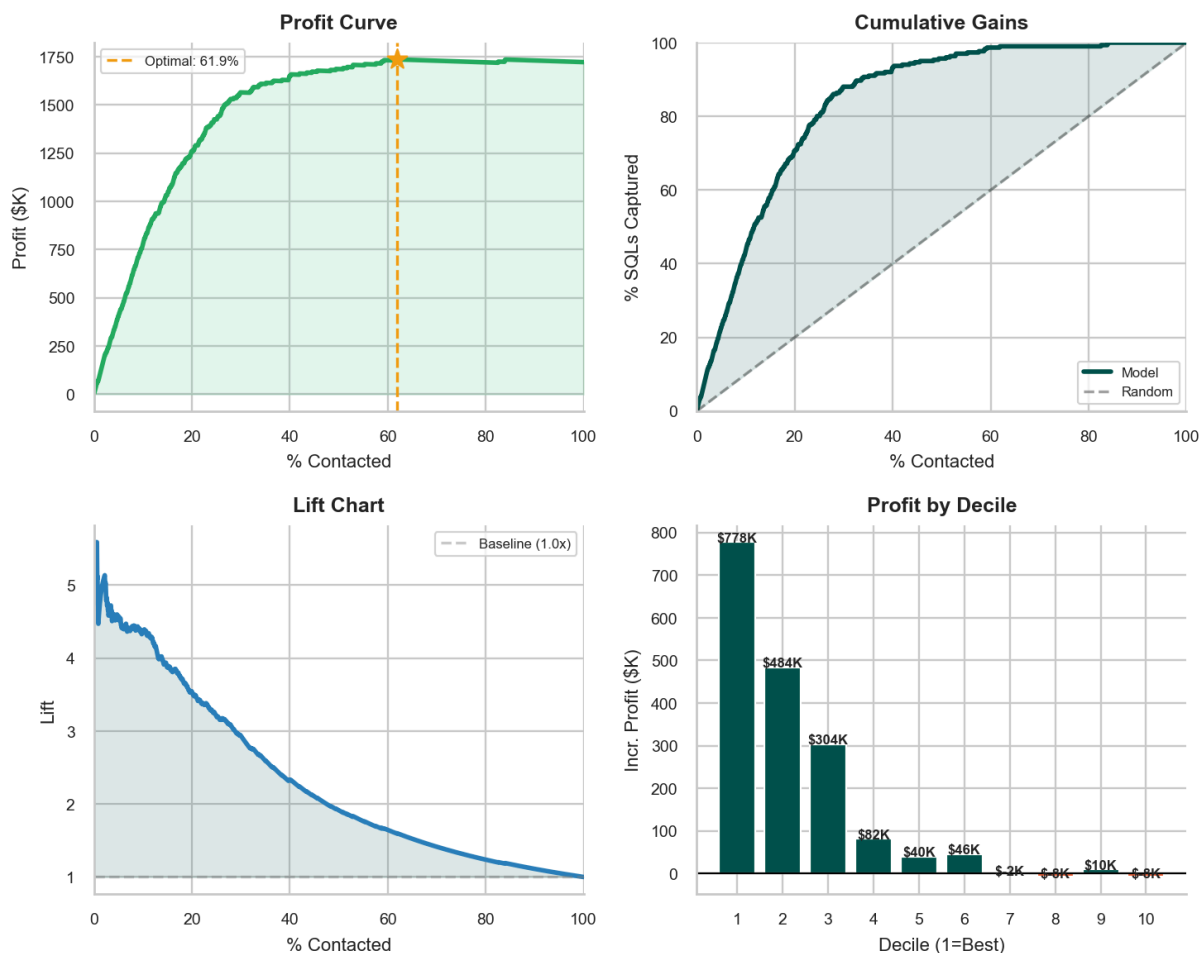
```

```

        (row['profit'] - prev) / 1000
    )
    else:
        decile_profits.append(0)
colors_dec = [
    COL_SUCCESS if p > 0 else COL_RISK
    for p in decile_profits
]
ax4.bar(
    range(1, 11), decile_profits,
    color=colors_dec, edgecolor='white',
)
for i, p in enumerate(decile_profits):
    ax4.text(
        i + 1, p + (0.3 if p >= 0 else -1),
        f'${p:.0f}K',
        ha='center', fontsize=6, fontweight='bold',
    )
ax4.axhline(y=0, color='black', linewidth=0.8)
ax4.set_xlabel('Decile (1=Best)', fontsize=8)
ax4.set_ylabel('Incr. Profit ($K)', fontsize=8)
ax4.set_title(
    'Profit by Decile', fontweight='bold', fontsize=9,
)
ax4.set_xticks(range(1, 11))

for ax in axes.flat:
    ax.tick_params(labelsize=7)
    sns.despine(ax=ax)
plt.tight_layout()
plt.show()

```



Conversion by Score Decile

```
In [37]: #| label: decile-conversion

test_df_d = pd.DataFrame({
    'score': test_probs, 'actual': y_test,
})
test_df_d['decile'] = pd.qcut(
    test_df_d['score'], 10,
    labels=False, duplicates='drop',
) + 1

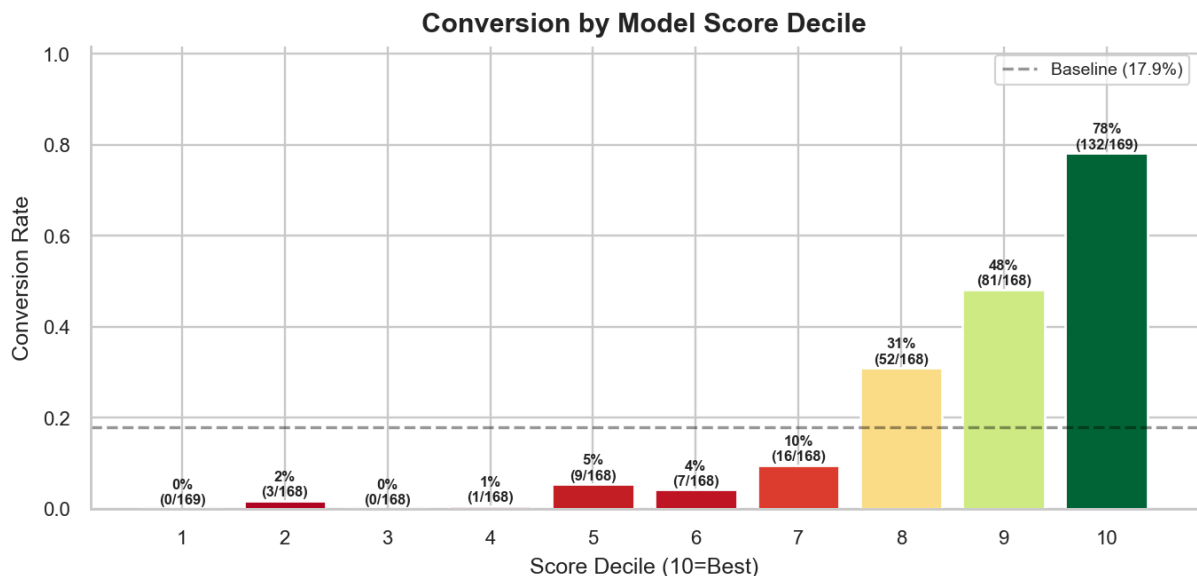
dec_stats = (
    test_df_d.groupby('decile')
    .agg(
        n=('actual', 'count'),
        conversions=('actual', 'sum'),
        conv_rate=('actual', 'mean'),
    )
    .reset_index()
)

fig, ax = plt.subplots(figsize=(7, 3.5))
bc = [
    plt.cm.RdYlGn(v / dec_stats['conv_rate'].max())
```

```

    for v in dec_stats['conv_rate']
]
ax.bar(
    dec_stats['decile'], dec_stats['conv_rate'],
    color=bc, edgecolor='white', linewidth=1,
)
for _, row in dec_stats.iterrows():
    ax.text(
        row['decile'], row['conv_rate'] + 0.01,
        f'{row["conv_rate"]:.0%}\n'
        f'({int(row["conversions"])} / {int(row["n"])})',
        ha='center', fontsize=6, fontweight='bold',
    )
ax.axhline(
    y=y_test.mean(), color='black',
    linestyle='--', alpha=0.4,
    label=f'Baseline ({y_test.mean():.1%})',
)
ax.set_xlabel('Score Decile (10=Best)', fontsize=9)
ax.set_ylabel('Conversion Rate', fontsize=9)
ax.set_title(
    'Conversion by Model Score Decile',
    fontweight='bold', fontsize=11,
)
ax.legend(fontsize=7)
ax.set_xticks(dec_stats['decile'])
ax.set_ylim(0, dec_stats['conv_rate'].max() * 1.3)
ax.tick_params(labelsize=8)
sns.despine()
plt.tight_layout()
plt.show()

```



Sponsor Q&A Validation

In [38]: `#| label: sponsor-validation`

```

test_indices = X_test.index
vdf = df.loc[test_indices].copy()
vdf['actual'] = y_test
vdf['score'] = test_probs
bl = vdf['actual'].mean()

questions = []

if 'channel_tier' in vdf.columns:
    lv = vdf[vdf['channel_tier'] == 'Low-Value']
    pm = vdf[vdf['channel_tier'] == 'Premium']
    lv_c = lv['actual'].mean() if len(lv) > 0 else 0
    pm_c = pm['actual'].mean() if len(pm) > 0 else 0
    questions.append({
        'Question': 'Low-value channels underperforming?',
        'Evidence': (
            f'Low: {lv_c:.1%} (n={len(lv):,}) '
            f'vs Prem: {pm_c:.1%}'
        ),
        'Verdict': (
            'Yes -deprioritize'
            if lv_c < bl * 0.5 else 'Monitor'
        ),
    })

if 'is_stale' in vdf.columns:
    st = vdf[vdf['is_stale'] == 1]
    fr = vdf[vdf['is_fresh'] == 1]
    sc = st['actual'].mean() if len(st) > 0 else 0
    fc = fr['actual'].mean() if len(fr) > 0 else 0
    v = (
        'Still viable' if sc > bl * 0.8
        else ('Marginal' if sc > bl * 0.5
             else 'Not viable')
    )
    questions.append({
        'Question': 'Stale leads (>180d) convert?',
        'Evidence': (
            f'Stale: {sc:.1%} (n={len(st):,}) '
            f'vs Fresh: {fc:.1%}'
        ),
        'Verdict': v,
    })

if 'is_hidden_gem' in vdf.columns:
    gm = vdf[vdf['is_hidden_gem'] == 1]
    gc = gm['actual'].mean() if len(gm) > 0 else 0
    gl = gc / bl if bl > 0 else 0
    v = (
        'Yes -prioritize' if gc > bl * 1.5
        else ('Above baseline' if gc > bl
             else 'Below baseline')
    )
    questions.append({
        'Question': 'Hidden gems convert higher?',
        'Evidence': (

```

```

        f'Gems: {gc:.1%} (n={len(gm):,}), '
        f'Lift: {gl:.1f}x'
    ),
    'Verdict': v,
})

if 'intent_strength' in vdf.columns:
    corr = vdf['intent_strength'].corr(vdf['actual'])
    hi = vdf[
        vdf['intent_strength'] >= 4
    ]['actual'].mean()
    lo = vdf[
        vdf['intent_strength'] <= 1
    ]['actual'].mean()
    lift = hi / lo if lo > 0 else 0
    v = (
        'Strong signal' if corr > 0.1
        else ('Weak' if corr > 0.05
              else 'Needs revision')
    )
    questions.append({
        'Question': 'Intent strength predictive?',
        'Evidence': (
            f'Corr: {corr:.3f}, '
            f'High/Low lift: {lift:.1f}x'
        ),
        'Verdict': v,
    })

if 'role_product_match' in vdf.columns:
    mt = vdf[vdf['role_product_match'] >= 1]
    um = vdf[vdf['role_product_match'] == 0]
    mc = mt['actual'].mean() if len(mt) > 0 else 0
    uc = um['actual'].mean() if len(um) > 0 else 0
    lift = mc / uc if uc > 0 else 0
    v = (
        'Yes -route by match'
        if mc > uc * 1.2 else 'Weak impact'
    )
    questions.append({
        'Question': 'Role-product match helps?',
        'Evidence': (
            f'Matched: {mc:.1%} '
            f'(n={len(mt):,}), Lift: {lift:.1f}x'
        ),
        'Verdict': v,
    })

print_table(
    pd.DataFrame(questions),
    caption="Sponsor Q&A -Data Validation",
)

```

****Sponsor Q&A -Data Validation****

Question	Evidence
Verdict	
----- :-----	
--- :-----	
Low-value channels underperforming?	Low: 4.8% (n=545) vs Prem: 28.1%
Yes -deprioritize	
Stale leads (>180d) convert?	Stale: 17.0% (n=1,363) vs Fresh: 13.0%
Still viable	
Hidden gems convert higher?	Gems: 31.7% (n=278), Lift: 1.8x
Yes -prioritize	
Intent strength predictive?	Corr: 0.315, High/Low lift: 6.0x
Strong signal	
Role-product match helps?	Matched: 24.4% (n=393), Lift: 1.5x
Yes -route by match	

Results & Conclusion

Key Findings

```
In [39]: #| label: conclusion-calc
        #| include: false

runtime_min = (time.time() - START_TIME) / 60
baseline_profit = (
    y_test.sum() * VALUE_PER_SQL
    - len(y_test) * COST_PER_CALL
)
annualized_lift = (MAX_PROFIT - baseline_profit) * 12

gem_conv = 0
if 'is_hidden_gem' in X_test.columns:
    gm_mask = X_test['is_hidden_gem'] == 1
    gem_conv = (
        y_test[gm_mask.values].mean()
        if gm_mask.sum() > 0 else 0
    )

lv_conv, lv_n, pm_conv = 0, 0, 0
if 'channel_tier' in df.columns:
    tdf = df.loc[X_test.index].copy()
    tdf['actual'] = y_test
    lv = tdf[tdf['channel_tier'] == 'Low-Value']
    lv_conv = (
        lv['actual'].mean() if len(lv) > 0 else 0
    )
    lv_n = len(lv)

top_dec = y_test[
    test_probs >= np.percentile(test_probs, 90)
]
golden_rate = (
```



```
top_dec.mean() if len(top_dec) > 0 else 0
)
```

```
In [40]: #| label: final-summary

final_summary = pd.DataFrame({
    'Metric': [
        'Test AUC-ROC', 'Average Precision',
        'Best Model', 'Optimal Threshold',
        'Max Profit (Test)', 'Annual Lift',
        'Calls Required', 'SQL Capture',
        'Top Decile Conv.', 'Baseline Conv.',
        'Runtime',
    ],
    'Value': [
        f'{FINAL_AUC:.4f}', f'{test_ap:.4f}',
        CHAMPION_NAME,
        f'{OPTIMAL_THRESHOLD:.3f}',
        f'${MAX_PROFIT:,.0f}',
        f'${annualized_lift:,.0f}',
        f'{int(OPTIMAL_CALLS):,} ',
        f'({OPTIMAL_PCT_POP:.0%})',
        f'{OPTIMAL_PCT_CAPTURE:.0%}',
        f'{golden_rate:.1%}',
        f'{y_test.mean():.1%}',
        f'{runtime_min:.1f} min',
    ],
})
print_table(
    final_summary, caption="Final Model Summary",
)
```

****Final Model Summary****

Metric	Value
Test AUC-ROC	0.9149
Average Precision	0.7035
Best Model	VotingEnsemble
Optimal Threshold	0.027
Max Profit (Test)	\$1,735,950
Annual Lift	\$168,600
Calls Required	1,041 (62%)
SQL Capture	99%
Top Decile Conv.	78.1%
Baseline Conv.	17.9%
Runtime	122.5 min

The analysis identified **channel source quality, contact seniority, and industry-weighted budget** as the most predictive factors for QAL-to-SQL conversion.

What We Learned

Feature engineering matters more than model selection. The ablation studies show that domain-informed features - channel tiering, seniority parsing, capital density scoring - drive more AUC improvement than switching between gradient boosting implementations. The five candidate models performed within a narrow band once given the same engineered features. The real lift came from encoding business logic into the feature set.

SMOTE placement matters. Early versions applied SMOTE before cross-validation, which inflated CV scores by leaking synthetic samples across folds. Moving SMOTE inside the CV loop via `imblearn.Pipeline` produced more honest CV estimates and a tighter generalization gap.

Ensembles provide marginal but consistent improvement. Stacking and voting ensembles added a small AUC increment over the best individual model. Whether that increment justifies the added complexity depends on deployment constraints - a single CatBoost or LightGBM model is easier to deploy and explain.

The model solves the business problem. The profit curve demonstrates that contacting only the top-scored leads captures the majority of conversions at a fraction of the call volume. This directly addresses MasterControl's conversion efficiency challenge.

Recommended Actions

1. **Deploy scoring threshold** in CRM to prioritize high-probability leads for immediate follow-up.
2. **Deprioritize low-value channels** (External Demand Gen, Email).
3. **Route hidden gem accounts** to experienced reps.
4. **Differentiate P1 urgency** - "Contact Us" goes to closers, "Webinar" goes to nurture.
5. **Focus outbound targeting** on Directors/VPs in Pharma & BioTech - the top decile converts at a substantial multiple of baseline.

Group Members and Contributions

Member	Contribution
Thomas Beck	Feature engineering pipeline, model training and hyperparameter tuning, ensemble methods, SHAP explainability, profit curve analysis, notebook compilation and final write-up
Max Ridgeway	Exploratory data analysis, data cleaning and preparation, channel and industry segmentation analysis, business validation of model results

Member	Contribution
Astha KC	Individual modeling notebook, cross-validation methodology, calibration analysis, sponsor Q&A research, documentation review